

WAGER: An Exact Decomposition of Model-Pair Score Gains on Benchmarks with Strong Label Priors

Anonymous authors

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Abstract

A scene graph generation method reports a ten-point gain in mean recall. What did it buy? On benchmarks whose labels are partly determined by a feature both compared models observe — the object-class pair in relation prediction, the class frequency in long-tailed recognition — such a gain mixes two improvements: the new model may fit the label frequencies *within* groups of that feature more closely, which a frequency table also achieves, or match its predictions to the individual cases they were made for. Mean recall, zero-shot recall and significance tests leave the two mixed. We decompose the gain exactly: score each model’s prediction for one test case against the label of a *different* case in the same group: averaging over relabellings gives the *transported gain*, and what remains is a *within-group covariance gain*. The identity is exact in finite samples, needs no fitted baseline, held-out fold or tuning parameter, and costs $O(NK)$ on cached predictions. The remainder is an order-two U-statistic, holds for every Bregman score, and admits cluster-robust intervals. On two released MOTIFS checkpoints evaluated with their authors’ code, a ten-point mean-recall improvement corresponds to a proper-score difference that is overwhelmingly transported. The unmodified estimator transfers to a frozen-CLIP variant and to long-tailed image and text classification, exposing compositions aggregate scores hide — including a model that loses on every such measure yet discriminates individual cases significantly better.

1 Introduction

A scene graph generation method reports a ten-point improvement in mean recall on Visual Genome. A long-tailed classifier reports two points of accuracy on CIFAR-100-LT. What did those gains buy?

The question is not rhetorical, because on benchmarks like these a model can raise its score without becoming any better at the cases it is scored on. In relation prediction, the ordered object-class pair already predicts the relation strongly: given (man, surfboard), the label is very often riding, and a lookup table of training frequencies is famously competitive (Zellers et al., 2018). A new model can therefore improve by estimating $\Pr(Y \mid \text{class pair})$ more accurately — a real improvement, but one that no more distinguishes *these two men on surfboards* than the frequency table did, and one that evaporates the moment the relation frequencies shift. Or it can improve by telling those two cases apart: putting higher probability on the correct label for the individual image, beyond what the class pair alone implies. The reported score is the sum of the two, and the field’s own metrics cannot separate them — not mean recall, not zero-shot recall, and not a significance test on the difference, because both components sit inside the same number.

The problem is general and the community knows it: VQA systems collapse under changed answer priors (Agrawal et al., 2018), and benchmark audits keep finding multimodal models exploiting non-visual shortcuts that inflate exactly this kind of aggregate score (Zhou et al., 2025). Reporting standards have responded with prior-aware slicings — frequency-stratified recall, zero-shot recall, rarity-stratified accuracy (Section 2.4) — but every one of them describes a *single* model. None attributes the gain between two submitted systems.

The customary audit builds a prior-only reference: fit $\widehat{\Pr}(Y \mid \phi)$ from training data, score it, and see how far the full models beat it (Zellers et al., 2018). As a diagnostic for one model that is useful; as an answer to the pairwise question it is indirect and noisy. The reference has to be estimated, it needs a smoothing rule

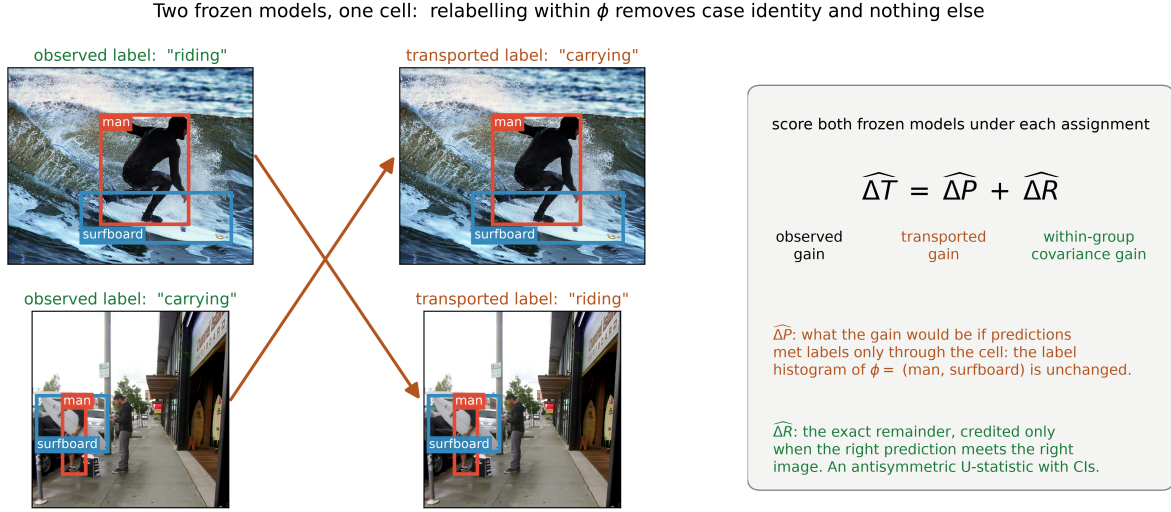


Figure 1: **The construction, on two real VG150 test relations.** Both share the group $\phi = (\text{man}, \text{surfboard})$ — the ordered object-class pair that already predicts the relation — yet carry different labels. Exchanging their labels (arrows) leaves the group and its label frequencies intact while breaking the link between each prediction and the image it was made for. Scoring two frozen models before and after the exchange splits their score difference exactly into a transported gain and a within-group covariance gain. Nothing about the construction is specific to relation prediction: it needs the two probability vectors, the labels, and the group identifiers.

for rare cells and a model family that may match neither compared system, and decomposing two models separately then subtracting gives no uncertainty for the difference — the quantity actually in dispute.

What this paper does. We decompose the score difference itself, exactly, with no fitted reference at all. The idea takes one sentence. Both models have already emitted a full probability vector for every test case, so each model’s score can be recomputed against *any* label, not only the one that was observed. Take two test cases in the same group — sharing the same value of a declared grouping variable ϕ , such as the object-class pair — and score each case’s predictions against the *other* case’s label. That relabelling preserves the group and its label frequencies while destroying the correspondence between a prediction and the case it was made for. Whatever score advantage survives is an advantage the new model holds over the group as a whole; whatever disappears was case-specific. Averaging within groups turns this into an identity,

$$\text{total score gain} = \underbrace{\text{transported gain}}_{\text{survives relabelling}} + \underbrace{\text{within-group covariance gain}}_{\text{destroyed by relabelling}},$$

exact in the observed sample, for any proper score, with no approximation, no held-out fold and no tuning parameter, at a cost of $O(NK)$ for N cases and K labels. Figure 1 shows the relabelling on two real Visual Genome cases. Appendix A works the whole construction through on four data points, in arithmetic a reader can check by hand, and uses the same four points to exhibit a model-selection decision the split gets right and the aggregate score gets wrong: a candidate whose entire gain survives relabelling becomes *worse* than the model it replaces once the deployment label frequencies shift, while one whose gain does not survive is untouched.

What the second component is The component destroyed by relabelling is not new, and naming it correctly predicts how it behaves. Written out, it equals the covariance, inside each group, between how much the new model moved its predicted probability for a label and whether that label occurred (Theorem 1) — the object isolated by the covariance rearrangement of a proper score going back to Yates (1982), so we call it the *within-group covariance gain*. It is close to, but not the same as, the *resolution* term of the

classical reliability–resolution partition (Murphy, 1973; DeGroot & Fienberg, 1983; Bröcker, 2009), and the difference matters in practice: classical resolution depends on a forecast only through the partition its level sets generate, so it survives any monotone rescaling of forecast values, whereas a covariance between those values does not. The two coincide exactly when both models are calibrated within groups. Section 4.1 measures the gap, and it is large enough that every comparison we report is between confidence-matched models — released checkpoints, which routinely differ sharply in calibration, cannot be audited on raw outputs.

We call the estimator WAGER; the acronym is a label, not a claim, and we do not expand it. Its two components are the *transported gain* ΔP — named for the operation that produces it, not for an interpretation of it — and the within-group covariance gain ΔR .

1.1 Contributions

1. **An audit of a released debiasing method.** On the two publicly released MOTIFS checkpoints of Tang et al. (2020), evaluated on the canonical VG150 PredCls split with the original authors’ code, TDE’s ten-point mean-recall improvement corresponds to a proper-score difference that is overwhelmingly transported, with a within-group covariance remainder at most a small fraction of it (Section 4.2).
2. **A drop-in tool for any closed-set benchmark.** The estimator needs four aligned arrays — two $N \times K$ probability matrices, the labels, the group identifiers — runs in $O(NK)$ on cached predictions with no refitting, no held-out fold and no tuning parameter, and returns both components with cluster-robust intervals in one pass. We apply it unchanged to relation prediction, a frozen-CLIP variant, and long-tailed image and text classification; only ϕ changes. Code, cached outputs and a verification script are released.
3. **An exact identity, with inference.** For any score the gain splits exactly, in population and finite sample, into a transported gain and a remainder that is an order-two U-statistic; for every Bregman score that remainder is a within-group covariance between the models’ predicted-probability difference and the label indicator, and a change constant within groups contributes exactly zero. The influence function of the difference gives image-cluster-robust intervals in one pass, the estimator is asymptotically normal with a consistent sandwich variance, and at a trivial ϕ it reduces to a clustered Diebold–Mariano test. Three further exact results remove guesswork about design choices: the in-sample plug-in is attenuated by exactly $(n_c - 1)/n_c$, merging groups shifts the estimand by an explicit between-group covariance of either sign, and the transported component splits into a mean-forecast-fit term minus a dispersion term — which is why we do not call it prior fit.
4. **Validation, and honest limits.** Simulations establish interval coverage under image clustering and delimit what the components do *not* separate: prior-only improvements stay out of the covariance channel, unrecorded within-group shortcuts enter it, and monotone recalibration moves score between the channels, which is why confidence matching is part of the procedure. Equally fine, fully identified partitions can also disagree without limit, so declaring ϕ is a modelling decision.

1.2 Terminology

Table 4 in Appendix M fixes the paper’s vocabulary against standard counterparts. Two entries carry a caveat rather than a synonym, because no standard term matches exactly: ΔP is named for the operation that produces it, and ΔR is a covariance rearrangement’s term rather than the classical resolution it is often mistaken for. We use “group” and “cell” interchangeably.

1.3 Scope, and what is not claimed

The *identity* needs four aligned arrays: two $N \times K$ probability matrices, the labels, and the cell identifiers. Any closed-set probabilistic prediction task supplies them; free-form or generative evaluation, where no K -way vector exists, does not. The *procedure* we recommend needs cluster identifiers for the intervals

and, whenever the models differ visibly in confidence, a temperature fitted per model on held-out clusters (Section 4.1).

Four limits are worth stating before any result, and Section 5 returns to each. Everything is conditional on the declared ϕ : a positive ΔR says the new model predicts individual cases better *among cases sharing that feature*, and nothing about causality or robustness. Second, ΔR credits any signal varying within a cell, including an unrecorded shortcut; Proposition 3 bounds that exposure rather than removing it, and for our own Visual Genome pair a covariate correlated at 0.061 would account for the entire reported gain (Appendix J). Third, and this is the limitation we would most want carried away, ΔR is *not invariant to recalibration*: rescaling a fixed model’s confidence changes it while changing no ranking, no top-1 decision and no information, and in simulation a pure recalibration moves it by -0.488 , tens of times any real gain reported here. Every result below is for confidence-matched pairs. Fourth, a transported gain is not illegitimate — fitting the deployment label distribution better is real skill when that distribution is the one that matters — and it is also not purely prior fit (Proposition 1).

Organization. Appendix A works a four-point example and uses it to exhibit a decision the split changes. Section 2 places the construction in the literature on proper-score decomposition and forecast comparison. Section 3 develops the estimator and its theory. Section 4 validates it and audits two released checkpoints. Sections 5 and 6 discuss limits and conclude. Proofs are in Appendix F; three further empirical studies are in Appendix O.

2 Related work

The construction sits between two literatures that rarely meet: the decomposition of proper scores, which supplies the object being estimated, and the comparison of forecasters, which supplies the inferential target. We take those first, then the applied literature that motivates the problem.

2.1 Relation to classical proper-score decompositions

Conditional on a chosen grouping variable, a proper score splits into a calibration (reliability) term and a resolution term (Murphy, 1973; DeGroot & Fienberg, 1983), made conditioning-set explicit by Bröcker (2009), whose treatment already covers general strictly proper scores and multicategory outcomes.

Our ΔR is related to that resolution term but is not equal to it, and the distinction matters before anything is built on it. The representation of a resolution-like quantity as a *covariance* between forecast and outcome indicator is due to Yates (1982), whose covariance decomposition of the Brier score is the direct ancestor of Eq. (7). The classical resolution term of DeGroot & Fienberg (1983) is a different object: it depends on a forecast only through the partition generated by its level sets, so it is invariant under any strictly monotone relabelling of forecast values. A covariance between forecast values and outcomes has no such invariance. The two coincide — ΔR_c equals twice a difference of classical resolution terms — exactly when both models are calibrated within each cell of ϕ , and otherwise differ by a reliability–resolution cross term that a monotone recalibration moves. Section 4.1 measures the gap: a pure recalibration of one model, which cannot change either model’s classical resolution at all, moves $\widehat{\Delta R}$ by -0.488 and reverses the sign verdict of the audit in Section 4.2. We therefore name ΔR for the covariance it is, and adopt confidence matching as procedure rather than caveat.

What the classical decomposition does not supply is inference for a pairwise contrast. The point estimates do combine — both models share the ϕ -partition, so the same $(n_c - 1)/n_c$ attenuation applies to each, and subtracting two naive plug-ins recovers exactly the biased contrast — but inference does not. Variance estimators for the single-model decomposition (Siegert, 2014) target one resolution term in isolation and supply no covariance between it and a second model’s estimate from the same data. Rebuilding that from the joint law is routine rather than open (Ferro, 2007; DeLong et al., 1988). What is new is not that a contrast can be given a variance, but that it admits an *exact finite-sample decomposition* whose remainder is an antisymmetric U-statistic with a one-pass influence function, so the split and its uncertainty come from one construction.

2.2 Comparing two forecasters

Inference on a paired score differential is the classical territory of comparative forecast evaluation: the Diebold–Mariano test (Diebold & Mariano, 1995) and conditional predictive ability tests (Giacomini & White, 2006) ask whether one forecaster beats another, optionally conditional on covariates. That is the closest existing relative of a gain evaluated conditional on ϕ , and Corollary 3 makes the relation exact: at the trivial one-cell partition our total-gain statistic and its interval *are* a clustered Diebold–Mariano test, corrected for cluster rather than serial dependence. The decomposition is therefore best read as everything those tests already deliver plus a split that neither provides. A structurally close precedent lies outside forecast evaluation: DeLong et al. (1988) compare two correlated ROC curves’ areas through a generalized U-statistic covariance estimator — the same construction applied to an unconditional ranking statistic rather than a declared-cell gain.

Evaluating calibration within cells of a declared grouping, uniformly over many candidate groupings, is the subject of multicalibration (Hébert-Johnson et al., 2018), the formal treatment of the multiplicity question the choice of ϕ raises (Section 5).

2.3 Transport, permutation, and U-statistics

Strictly proper scoring rules reward honest predictive distributions (Gneiting & Raftery, 2007), and permutation methods have long been used to assess classifier performance (Ojala & Garriga, 2010). Conditional permutation tests generally estimate or assume a conditional distribution when the conditioning variables are continuous or high-dimensional. The setting here is different in a way that buys exactness: a benchmark supplies a *discrete* grouping variable, and the target is not generic conditional independence but one specific model-to-model score gain, so within-cell transport needs no nuisance law at all. The remainder is a U-statistic of order two in the sense of Hoeffding (1948), and its Hájek projection is what supplies the influence function of Section 3.9. The new element is the gain-contrast transport identity and the attribution it licenses, not permutation or U-statistic theory by itself.

Two further connections are worth naming. The transported component is what label-shift procedures manipulate post hoc: re-weighting a classifier’s outputs towards a new label distribution (Saerens et al., 2002; Lipton et al., 2018) can add or remove transported gain without touching case-level evidence, which Section 4.4 exploits as a falsification test, and which is why Appendix A’s model-selection reversal is a label-shift argument. And what an unrecorded grouping variable can hide is the subject of omitted-variable sensitivity analysis (Rosenbaum, 2002); Proposition 3 adapts that style of argument to bound how far an unrecorded covariate could move ΔR .

Usable-information analyses measure information accessible to a chosen predictor family; their central objects are model- or family-level quantities, whereas ours is narrower and operational, attributing the score difference between two concrete frozen models. A positive ΔR establishes nothing about causal or compositional structure.

2.4 Prior-aware evaluation in vision

Dataset bias has long complicated comparisons between recognition systems (Geirhos et al., 2020). Scene graph generation was posed as structured prediction over a whole image and solved by iterative message passing between object and relation nodes (Xu et al., 2017); against that formulation a frequency predictor of $p(\text{predicate} \mid \text{subject}, \text{object})$ is surprisingly competitive (Zellers et al., 2018); mean recall and causal interventions were proposed to reduce that dependence (Tang et al., 2020), though metric pathologies remain (Li et al., 2022) and the same diagnosis recurs in open-vocabulary and knowledge-enhanced work (Li et al., 2024). In visual question answering, GQA-OOD stratifies by answer rarity within question groups (Kervadec et al., 2021) and VQA-CP changes language priors between train and test (Agrawal et al., 2018); the shortcut is still targeted directly (Kuang et al., 2024), and benchmark-level audits of exploitable non-visual shortcuts have widened to multimodal evaluation generally (Zhou et al., 2025). In long-tailed recognition a classifier can raise aggregate accuracy by fitting training class frequencies more precisely without discriminating better

within a class (Duan et al., 2024); the canonical remedies rebalance the objective or the classifier (Cui et al., 2019; Cao et al., 2019; Menon et al., 2021), with a continuing stream of variants (Luo et al., 2025).

Reporting standards have responded to the shortcut: mean recall alongside recall, often broken down by predicate frequency (Tang et al., 2020; Li et al., 2022), and zero-shot recall (Lu et al., 2016) checking whether a gain survives on triplets absent from training. These are effective prior-aware slicings, splits and stress tests of a *single* model’s performance. What none provides is what we target: an exact decomposition, with uncertainty, of the score difference between two fixed systems. A stratified or zero-shot metric can show *that* a gain is uneven across the prior; it does not attribute the pairwise gain itself, on the full evaluation set, with a finite-sample identity and inference. The remedies, equally, intervene on how a model is trained rather than on how a trained pair is compared.

3 The estimator and its properties

This section states the construction of Section 1 formally and establishes its properties. The order is: notation and the object being decomposed (Section 3.1); the population identity and the covariance form of its remainder (Section 3.2), which Section 3.4 shows is not specific to the quadratic score; what happens when the grouping variable is made coarser (Section 3.5) or is missing a relevant covariate altogether (Appendix E.1); the finite-sample identity and its $O(NK)$ computation (Sections 3.6–3.7); why the counterfactual leaves each case out rather than fitting the group’s own frequencies (Section 3.8); and inference, both finite-sample and asymptotic (Sections 3.9–3.10). Proofs are in Appendix F. A reader who wants only what is needed to use the estimator can read Sections 3.1, 3.2, 3.6 and 3.9 and skip the rest.

3.1 Setup: what is being decomposed

Let $q_0(\cdot | x)$ and $q_1(\cdot | x)$ be the predictive distributions of an old and a new frozen model over K labels. Let $S(q, y)$ be a bounded proper score, oriented so that larger is better, and let $C = \phi(X)$ be a declared discrete grouping variable. In the relation-prediction example of Section 1, C is the ordered subject–object class pair; in a long-tailed classification benchmark it is a coarse superclass. We call a value of C a *cell* and require it to be declared before the audit is run. Define the instance-specific *gain vector*

$$H_x(y) = S(q_1(\cdot | x), y) - S(q_0(\cdot | x), y), \quad y = 1, \dots, K. \quad (1)$$

Unlike the realized gain $H_X(Y)$, this vector can be evaluated at a counterfactual label without rerunning either model.

Our default is the quadratic score

$$S_B(q, y) = 2q_y - \|q\|_2^2, \quad (2)$$

which differs from negative Brier loss only by a label-only constant. It is strictly proper and bounded, avoids probability clipping, and uses the entire predictive distribution. A floored log score is a sensitivity analysis.

3.2 Within-cell counterfactual transport

For a cell c , define three population quantities:

$$\Delta T_c = \mathbb{E}[H_X(Y) | C = c], \quad (3)$$

$$\Delta P_c = \mathbb{E}_{H|c} \mathbb{E}_{Y'|c}[H_X(Y')], \quad (4)$$

$$\Delta R_c = \Delta T_c - \Delta P_c. \quad (5)$$

Here Y' is an independent label drawn from the same cell. Thus ΔT_c is the observed score gain, ΔP_c is the gain retained after breaking the instance-specific prediction–label assignment while preserving the cell, and ΔR_c is the covariance gain lost by that transport.

Theorem 1 (Transport and covariance identities). *For any integrable score contrast H ,*

$$\Delta T_c = \Delta P_c + \Delta R_c, \quad \Delta R_c = \sum_{y=1}^K \text{Cov}(H_X(y), \mathbb{1}\{Y = y\} | C = c). \quad (6)$$

For the quadratic score, writing $\Delta q_y(X) = q_1(y | X) - q_0(y | X)$,

$$\Delta R_c = 2 \sum_{y=1}^K \text{Cov}(\Delta q_y(X), \mathbb{1}\{Y = y\} | C = c). \quad (7)$$

Consequently, if the gain vector is a function of c alone, then $\Delta R_c = 0$.

Equation (7) is the operational meaning of the construction: the new model receives covariance credit only to the extent that its probability change aligns more strongly with the label of the correct case within the same cell. Cell-constant score improvements, including a better fit to the cell's label histogram, remain in ΔP . A nonzero covariance does not assert causal or compositional understanding; unrecorded within-cell shortcuts contribute to it too, so ϕ must be declared and stress-tested.

3.3 What the transported gain measures

The surviving component is easy to misname. It is tempting to call ΔP a prior-fit gain, since transport replaces a case's label by an independent draw from its cell; earlier drafts of this paper did. The name is wrong.

Proposition 1 (Composition of the transported gain). *Under the quadratic score, write $\bar{q}_m(c) = \mathbb{E}[q_m(\cdot | X) | C = c]$ for model m 's mean forecast in cell c , $\text{Var}(q_m | c)$ for the within-cell covariance matrix of its forecast vector, and $p(c)$ for the cell's transport law. Then*

$$\Delta P_c = \underbrace{\left[\|p(c) - \bar{q}_0(c)\|^2 - \|p(c) - \bar{q}_1(c)\|^2 \right]}_{\text{improvement in mean-forecast fit}} - \underbrace{\left[\text{tr Var}(q_1 | c) - \text{tr Var}(q_0 | c) \right]}_{\text{increase in within-cell dispersion}}. \quad (8)$$

Proved in Appendix F. In words: under the quadratic score ΔP_c equals the improvement in how closely the new model's *mean* within-cell forecast sits to the cell's label frequencies, minus the increase in the within-cell dispersion of its forecasts. Only the first term is prior fit. The second rewards the new model for making less dispersed predictions at a fixed mean forecast, so ΔP can absorb an entire observed gain with both models' mean forecasts sitting exactly on the cell frequencies — the four-point counterexample of Appendix N does just that. This is the reliability-and-sharpness channel of the classical partition, not a prior-fitting channel, which is why we name it for the operation that produces it; it is also why confidence matching moves $\widehat{\Delta P}$ so sharply, since a temperature change is almost purely a change in dispersion.

3.4 Generalization: the Bregman-score family

Nothing in the proof of Theorem 1 uses properness: the identity and the covariance sum follow from the definitions alone, for any score. What is specific to the quadratic score is the further reduction to Eq. (7), in which the part of $H_x(y)$ not depending on y integrates out because $\sum_y \mathbb{1}\{Y = y\} = 1$ almost surely. The same cancellation occurs for every strictly proper score generated by a Bregman divergence, so Eq. (7) and the log-score identity used as a robustness check in Section 4.2 are two instances of one theorem. For a strictly convex, differentiable ψ on the simplex, the Bregman score is $S_\psi(q, y) = \psi(q) + \langle \nabla \psi(q), e_y - q \rangle$; taking $\psi(q) = \|q\|_2^2$ recovers the quadratic score and $\psi(q) = \sum_k q_k \log q_k$ the log score (Gneiting & Raftery, 2007).

Theorem 2 (Bregman-score covariance identity). *Let $H_x(y) = S_\psi(q_1(\cdot | x), y) - S_\psi(q_0(\cdot | x), y)$ for a Bregman score generated by a strictly convex, differentiable ψ , and write $\Delta g_y(x) = \partial_y \psi(q_1(\cdot | x)) - \partial_y \psi(q_0(\cdot | x))$ for the contrast of the y -th gradient coordinate. Then*

$$\Delta R_c = \sum_{y=1}^K \text{Cov}(\Delta g_y(X), \mathbb{1}\{Y = y\} | C = c). \quad (9)$$

Corollary 1 (Quadratic and log score). *For $\psi(q) = \|q\|_2^2$, $\Delta g_y(x) = 2\Delta q_y(x)$ and Eq. (9) is Eq. (7). For $\psi(q) = \sum_k q_k \log q_k$, $\Delta g_y(x) = \log q_1(y | x) - \log q_0(y | x)$, and*

$$\Delta R_c = \sum_{y=1}^K \text{Cov} \left(\log \frac{q_1(y | X)}{q_0(y | X)}, \mathbb{1}\{Y = y\} \mid C = c \right). \quad (10)$$

Both are proved in Appendix F: the remainder equals the same within-cell covariance sum with the probability difference replaced by the contrast of the y -th gradient coordinate of ψ . Two named cases follow. For $\psi(q) = \|q\|_2^2$ the gradient contrast is $2\Delta q_y(x)$ and the identity is Eq. (7); for the negative entropy it is the log-likelihood-ratio contrast $\log q_1(y | x) - \log q_0(y | x)$, so the log-score check of Section 4.2 carries the same exact covariance meaning rather than being merely a sensitivity analysis. The implemented log score floors each probability at $\epsilon = 10^{-6}$ to avoid $-\infty$ on an unseen label; the identity is exact for the unfloored score and the floored version is a check on it, negligibly different whenever no relevant probability is within ϵ of zero.

3.5 Coarsening the cell, and an unrecorded covariate

Two questions about the declared ϕ have exact answers. First, what happens when ϕ is made coarser.

Proposition 2 (Coarsening decomposition). *Let $\bar{\phi} = g(\phi)$ be any coarsening of the grouping variable, so that each cell $\bar{c}$ of $\bar{\phi}$ is a union of cells of ϕ . Then*

$$\Delta R_{\bar{c}} = \mathbb{E}[\Delta R_C \mid \bar{\phi} = \bar{c}] + \sum_{y=1}^K \text{Cov}(\mathbb{E}[H_X(y) \mid C], \Pr(Y = y \mid C) \mid \bar{\phi} = \bar{c}). \quad (11)$$

Proved in Appendix F. Merging cells changes the estimand by an explicit between-cell covariance term, taken over the finer cells nested inside each coarse one, between each fine cell’s typical gain at a label and that label’s frequency there. The term is not sign definite, so a coarser grouping can inflate or deflate the credited covariance gain rather than leaving it alone, and it vanishes only when one of the two quantities is constant within each coarse cell. This is why the finest defensible ϕ is the conservative default, and Section 4.5 reports the empirically inflating case.

Second, read in the other direction, the same identity says what happens when a relevant covariate is *never* declared — the estimator’s main scope limit. Writing Z for the unrecorded variable and (ϕ, Z) for the fully adjusted cell, the bias of using ϕ alone is exactly that between-cell covariance term with (ϕ, Z) as the fine partition. It is exact but needs Z ’s joint law with the gain and the label. The following replaces that with a single elicited correlation $\bar{\rho}$ between an unrecorded variable’s effect on the within-cell gain and its effect on the within-cell label distribution, in the spirit of the robustness value that Cinelli & Hazlett (2020) elicit for regression confounding.

Proposition 3 (Sensitivity bound for an unrecorded confounder). *Suppose $H_x(y) \in [-M, M]$ for all x, y (WAGER already requires a bounded score). Fix a cell c of ϕ , write $a_y = \mathbb{E}[H_X(y) \mid \phi, Z]$, $b_y = \Pr(Y = y \mid \phi, Z)$ (both conditional on $\phi = c$) for the two K -vectors on the right of Eq. (11), and let*

$$\rho = \frac{\sum_{y=1}^K \text{Cov}(a_y, b_y \mid c)}{\sqrt{\sum_{y=1}^K \text{Var}(a_y \mid c)} \sqrt{\sum_{y=1}^K \text{Var}(b_y \mid c)}} \in [-1, 1] \quad (12)$$

be their aggregate correlation. Then

$$\begin{aligned} |\Delta R_c(\phi) - \mathbb{E}[\Delta R_{(\phi, Z)} \mid \phi = c]| &\leq |\rho| \sqrt{\sum_{y=1}^K \text{Var}(a_y \mid c)} \sqrt{\sum_{y=1}^K \text{Var}(b_y \mid c)} \\ &\leq |\rho| \sqrt{\sum_{y=1}^K \text{Var}(H_X(y) \mid \phi = c)} \sqrt{\sum_{y=1}^K p_y(c)(1 - p_y(c))}, \end{aligned} \quad (13)$$

where $p_y(c) = \Pr(Y = y \mid \phi = c)$; the second inequality holds termwise ($\text{Var}(a_y \mid c) \leq \text{Var}(H_X(y) \mid c)$ and $\text{Var}(b_y \mid c) \leq p_y(c)(1 - p_y(c))$ by the law of total variance, since conditioning on more variables cannot

increase the variance of a conditional expectation) followed by monotonicity of the square root applied to each sum separately – a product of two square-root sums, not a sum of per-label square roots, which the law of total variance alone would not license.

Corollary 2 (Free bound). *Each a_y ranges over $[-M, M]$ as a conditional expectation of H , so $\text{Var}(a_y | c) \leq M^2$ and $\sum_y \text{Var}(a_y | c) \leq KM^2$. The label probabilities satisfy $\sum_y b_y = 1$, which gives $\sum_y b_y^2 \leq 1$ and, by Cauchy-Schwarz, $\sum_y (\mathbb{E}b_y)^2 \geq 1/K$; hence*

$$\sum_{y=1}^K \text{Var}(b_y | c) = \mathbb{E}\left[\sum_y b_y^2\right] - \sum_y (\mathbb{E}b_y)^2 \leq 1 - \frac{1}{K}. \quad (14)$$

Substituting both into Eq. (13) gives the data-free bound

$$|\Delta R_c(\phi) - \mathbb{E}[\Delta R_{(\phi, Z)} | \phi = c]| \leq |\rho| M \sqrt{K - 1}. \quad (15)$$

Both are proved in Appendix F: the bias is at most $\bar{\rho}$ times a product of two dispersion factors, with a data-free form $\bar{\rho}M\sqrt{K-1}$ needing only the label cardinality and the score bound. Because $\bar{\rho}$ is a correlation rather than a domain-specific odds ratio it can be elicited the same way across benchmarks. Both forms are evaluated on our own estimates there, and the sharp one is the less comfortable: an unrecorded covariate correlated at 0.061 would account for the entire covariance gain we report in Section 4.3.

3.6 Exact finite-sample decomposition

Consider cell c with $n_c \geq 2$ examples. Write $H_i(y) = H_{x_i}(y)$. WAGER computes

$$\widehat{\Delta T}_c = \frac{1}{n_c} \sum_{i \in c} H_i(y_i), \quad (16)$$

$$\widehat{\Delta P}_c = \frac{1}{n_c(n_c - 1)} \sum_{i \neq j \in c} H_i(y_j), \quad (17)$$

$$\widehat{\Delta R}_c = \binom{n_c}{2}^{-1} \sum_{i < j \in c} A_{ij}, \quad (18)$$

where the antisymmetric kernel is

$$A_{ij} = \frac{1}{2} \{H_i(y_i) + H_j(y_j) - H_i(y_j) - H_j(y_i)\}. \quad (19)$$

Theorem 3 (Finite-sample WAGER identity). *For every cell with $n_c \geq 2$, without a distributional assumption,*

$$\widehat{\Delta T}_c = \widehat{\Delta P}_c + \widehat{\Delta R}_c. \quad (20)$$

Moreover, when the cell's examples are drawn i.i.d. from the cell population, $\widehat{\Delta R}_c$ is an unbiased order-two U -statistic for ΔR_c . Independence, not exchangeability alone, is what unbiasedness requires: the independent-coupling definition of ΔP_c pairs an example with a label drawn independently from the same cell, so $\mathbb{E}[H_1(Y_2)]$ matches it only for independent within-cell draws.

Proved in Appendix F. The additivity holds in the observed sample with no distributional assumption; only the unbiasedness needs one, and it needs independence rather than exchangeability alone, since the transported quantity pairs a case with a label drawn independently from the same cell.

Dataset-level quantities weight cells by their identified counts, $\widehat{\Delta Z} = N_*^{-1} \sum_{c: n_c \geq 2} n_c \widehat{\Delta Z}_c$ for $Z \in \{T, P, R\}$. Singleton cells cannot identify a within-cell counterfactual, so we exclude and report them rather than inventing a smoothed estimate. If $\widehat{\Delta T} > 0$ the descriptive share $\widehat{\Delta R}/\widehat{\Delta T}$ may exceed one, when covariance improves while the transported channel deteriorates, and it is undefined for a nonpositive total; we therefore report $\widehat{\Delta R}$ as the primary result rather than a thresholded ratio.

3.7 Linear-time computation

Naively evaluating Eq. (17) is quadratic in cell size. Let m_{cy} be the count of label y in cell c . For example $i \in c$, its transported score is

$$P_i = \frac{\sum_{y=1}^K m_{cy} H_i(y) - H_i(y_i)}{n_c - 1}. \quad (21)$$

Averaging P_i yields Eq. (17); averaging $H_i(y_i) - P_i$ yields Eq. (18). Thus the entire estimator costs $O(NK)$ time and $O(NK)$ storage for cached prediction vectors. No model fitting, projection fold, hierarchy, or smoothing hyperparameter appears.

3.8 Why leave-one-out transport, not a fitted-in-sample prior

Equation (21) excludes example i 's own label from its transported score. It is natural to ask what is lost by the simpler alternative of fitting the cell's empirical label frequency on all n_c examples, including i , and using that fitted frequency as the counterfactual, i.e. treating $\hat{p}_c(y) = m_{cy}/n_c$ as a prior model and setting $\tilde{P}_i = \sum_y \hat{p}_c(y) H_i(y) = n_c^{-1} \sum_y m_{cy} H_i(y)$. This is the natural plug-in estimator, and it is what a frequency-collapsed projection baseline computes when no separate evaluation fold is withheld from fitting.

Proposition 4 (Exact attenuation of the in-sample plug-in). *For every cell with $n_c \geq 2$ and every $i \in c$, writing $\hat{P}_i$ for Eq. (21) and $\hat{R}_i = H_i(y_i) - \hat{P}_i$,*

$$\tilde{P}_i = \hat{P}_i + \frac{1}{n_c} \hat{R}_i, \quad \tilde{R}_i := H_i(y_i) - \tilde{P}_i = \frac{n_c - 1}{n_c} \hat{R}_i. \quad (22)$$

Consequently the in-sample plug-in's dataset-level statistic obeys $\widetilde{\Delta R} = N_^{-1} \sum_{c: n_c \geq 2} (n_c - 1) \widehat{\Delta R}_c$, a multiplicative shrinkage of every cell's leave-one-out covariance gain by the factor $(n_c - 1)/n_c$.*

The shrinkage is deterministic, not a variance statement: whenever $\widehat{\Delta R}_c \neq 0$ the in-sample plug-in is biased for it at every finite n_c , worst in the smallest identified cells (50% at $n_c = 2$). The standard remedy for such self-prediction bias is a projection/evaluation split, which restores unbiasedness by discarding examples from both stages and introduces a split-fraction hyperparameter and typically higher variance. Leave-one-out transport removes the bias exactly, at the full sample, for every cell with $n_c \geq 2$, with no fitted nuisance distribution at all.

3.9 Inference

For point estimation we use all unordered within-cell pairs. For uncertainty, define $\bar{A}_i = (n_c - 1)^{-1} \sum_{j \neq i} A_{ij}$ and $\widehat{\Delta R}_c = n_c^{-1} \sum_i \bar{A}_i$. The estimated Hájek influence contribution for random cell membership is

$$\hat{\psi}_i = 2\bar{A}_i - \widehat{\Delta R}_c - \widehat{\Delta R}. \quad (23)$$

We sum $\hat{\psi}_i$ within image and use the resulting one-way sandwich variance, so multiple relations from the same image are not treated as independent. Total-gain influence is $H_i(y_i) - \widehat{\Delta T}$; prior-gain and share intervals follow from the exact identity and delta method.

As a complementary finite-group diagnostic we apply an independently sampled cyclic shift to the labels inside every cell, preserving cell size and label histogram, and recompute $\widehat{\Delta R}$ under B shifts to obtain a one-sided p -value; Appendix H gives the algorithm. It is exact under the sharp null that labels are exchangeable relative to the frozen model outputs within each cell. Because relations share images, our primary real-data uncertainty statement is the image-cluster-robust interval and the randomization test is a secondary relation-level diagnostic.

3.10 Asymptotic normality

The influence function and sandwich variance behind the reported intervals are derived in Appendix G through a Hájek-projection expansion, the standard route for order-two U-statistics (Serfling, 1980; van der

Vaart, 1998). The following supplies the limit theorem, under conditions that hold whenever the score is bounded and no cluster dominates the sample.

Theorem 4 (Asymptotic normality of the covariance-gain estimator). *Suppose (i) $H_x(y) \in [-M, M]$ for all x, y ; (ii) examples are partitioned into clusters $g = 1, \dots, G$ (e.g. images) that are mutually independent, with cluster sizes having a finite third moment and $\max_g |g| = o(\sqrt{G})$; (iii) the limiting variance $\sigma^2 = \lim_{G \rightarrow \infty} \text{Var}(\sqrt{N_*}(\widehat{\Delta R} - \theta))$ exists and is strictly positive, where θ is the identified-subpopulation estimand of Section G; and (iv) ϕ takes values in a fixed, finite set, each with strictly positive limiting population mass, so that every eligible cell’s size n_c diverges almost surely as $N \rightarrow \infty$. Then*

$$\sqrt{N_*}(\widehat{\Delta R} - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2), \quad (24)$$

and the plug-in sandwich variance $\widehat{\sigma}^2 = N_* \frac{G}{G-1} \sum_g (\sum_{i \in g} \widehat{\psi}_i / N_*)^2$ of Section G is consistent for σ^2 .

Proved in Appendix F; Appendix K discusses what condition (iv) costs and states the additional condition on the number of cells that the dataset-level remainder requires. That condition is an idealisation no heavy-tailed benchmark attains in sample, which is why Section 4.1’s coverage simulation, not this theorem, is what licenses reading the reported intervals at finite N .

Relation to comparative forecast evaluation.

Corollary 3 (Relation to Diebold–Mariano/Giacomini–White). *Under the trivial partition $\phi \equiv \text{const}$ (one cell containing every example), $\widehat{\Delta T} = N^{-1} \sum_i H_i(y_i)$ is the sample mean of the paired loss differential between the two models, and Theorem 4 applied to $\widehat{\Delta T}$ with its image-clustered interval is the grouped, cluster-robust form of the Diebold–Mariano statistic for that differential (Diebold & Mariano, 1995); because the conditioning information set is empty, it is equivalently a degenerate instance of the Giacomini–White conditional predictive-ability test (Giacomini & White, 2006).*

The corollary is immediate from the definitions: DM and GW test whether a paired loss differential’s mean is zero, and $\widehat{\Delta T}$ at the trivial partition is exactly that differential’s sample mean. Neither test forms $\widehat{\Delta P}$ or $\widehat{\Delta R}$; both target ΔT and stop. One qualification on how far the reading travels: the corollary is stated at the trivial partition, whereas the totals reported at a nontrivial ϕ are N_* -weighted means over the *identified* subsample. Because ΔT does not depend on ϕ except through that exclusion, each reported $\widehat{\Delta T}$ is a clustered DM statistic for the paired differential on its identified subsample, which at the coverages here (99.0% on VG150, 98.9% in Section 4.2) differs from the full-sample statistic only in the excluded singletons.

4 Experiments

Section 4.1 validates the instrument under controlled alternatives where the truth is known by construction, and establishes what it fails to separate. Section 4.2 is the paper’s main result: an audit of two publicly released checkpoints of a widely cited debiasing method, chosen so that the audited models are not ours and the claimed improvement is one the field already reports. Sections 4.3–4.4 then return to relation prediction with predictors built so that their information content is controlled, which is the complementary evidence a released checkpoint cannot give: there the ground truth about what each model can see is known, so the decomposition can be checked against it. Two further studies — long-tailed image classification on CIFAR-100-LT across seeds and imbalance ratios, and long-tailed text classification on 20-Newsgroups-LT — apply the identical estimator in other data modalities and are reported in Appendix O, together with the granularity, score and minimum-cell-size sensitivity analyses. All confidence intervals are 95%.

4.1 Controlled validation

We generate 24 cells over five labels, drawing each cell’s label distribution from a Dirichlet law and taking the old model to be exactly the cell prior. The new model interpolates towards an oracle, $q_1 = (1 - \beta)q_0 + \beta q_{\text{oracle}}$ with q_{oracle} placing 0.98 on the realized label; forty repetitions run at each $\beta \in \{0, .1, \dots, .8\}$. The estimated

covariance gain grows linearly and monotonically with the injected signal. Type-I behaviour is assessed by perturbing the prior predictions with mean-zero noise and reassigning the perturbed vectors within each cell, which guarantees no systematic prediction–label alignment survives: over 200 such datasets the 99-shift randomization test rejects in 0.030 of runs at level 0.05, with a mean estimate of -8.97×10^{-5} and approximately uniform null p -values.

Coverage of the primary interval. The paper’s primary uncertainty statement is the image-clustered Hájek interval, so we validate it in the regime that stresses it: relations grouped into images (one to about thirty each) whose oracle weight shares a latent per-image effect, with Zipf-distributed cell sizes in which most identified cells hold exactly two examples. Against the design expectation from 4,000 replications, the cluster-robust interval covers in 94.0% of 500 runs while an interval treating relations as independent covers in only 88.6% — quantifying both the mild finite-sample undercoverage and the necessity of clustering.

Discriminant validity Three designs test that the channels separate what they claim to. A *calibration-only* alternative, where the new model is a temperature-softened copy of an overconfident informative model so the pair carries identical case information, is credited raw with a large negative covariance gain (mean -0.488), which the protocol below reduces to -0.0003 (sd 0.0017 over 200 runs). A *prior-only* improvement, where the new model corrects a heteroscedastically noisy estimate of the cell prior, yields a covariance gain centred at zero (0.0002, sd 0.0035) with a clearly positive transported channel. And an *unrecorded within-cell shortcut* — a binary feature varying inside each cell that shifts the label distribution and is used by the new model — is credited to covariance ($+0.041$, sd 0.004): $\widehat{\Delta R}$ measures all within-cell predictive signal relative to the declared ϕ , shortcut or otherwise.

The calibration-matched protocol. The first of those designs identifies a limitation serious enough to be part of the procedure rather than a caveat about it: because ΔR is a covariance between probability movements and labels (Theorem 1), rescaling a fixed model’s confidence moves score between the two components while changing no ranking, no decision and no case-level information. We therefore adopt the following control wherever the compared models differ visibly in confidence. Split the evaluation units — images, where examples are clustered — at random into halves. On the first half, fit each model a single temperature T by maximum held-out likelihood on $q^{1/T}$ renormalized. On the second half, apply each model’s own fitted temperature and run the decomposition there. The protocol costs nothing but a data split, it uses no label information from the audited half, and it is what the simulation above shows removes the confidence-scaling channel almost exactly.

How many splits each study averages over differs, and the difference is worth stating rather than glossing. The long-tailed classification study of Appendix O.5 averages over twenty random splits. The audit of Section 4.2 and the matched rows of Section 4.4 each use a single split at a fixed seed. For those two the split is a source of variability the reported interval does not include, and it is not negligible: the calibration-only arm above has a between-split standard deviation of 0.001725, exceeding the half-width of the audit’s own reported interval (0.00115). The audit’s conclusion does not turn on this — its estimate is indistinguishable from zero and stays so under any split — but a reader comparing interval widths across our tables should know that two of the three matched studies report within-split uncertainty only. It does not make ΔR invariant to recalibration in general; it fixes the confidence scale at which the comparison is made, which is what a comparison of two *fixed* models needs.

4.2 Auditing a published debiasing method

The question the decomposition exists to answer concerns systems other people release and the field already compares, so the primary application is to a published pair. Tang et al. (2020) release a causal MOTIFS PredCIs checkpoint that yields two models from one set of weights: the biased baseline (MOTIFS-SUM) and its Total Direct Effect counterpart (MOTIFS-TDE), which subtracts a context-only counterfactual at inference. TDE is among the most cited debiasing methods in scene graph generation and is reported through mean recall, so it is exactly the kind of claimed improvement whose composition is at issue.

We evaluate both variants with the authors’ own code on the canonical VG150 test split, which fixes the vocabulary, split and relation set and makes the two prediction sets aligned relation-for-relation. Our reproduction matches the published behaviour: the baseline reaches $R@50 = 0.6612$ with $mR@50 = 0.1459$, and TDE trades the former for the latter, reaching $mR@50 = 0.2476$ and raising zero-shot recall to $zR@50 = 0.1432$. Top-1 predicate accuracy falls from 0.6981 to 0.5577. The audit uses 183,639 relations over 26,446 images and 7,127 class-pair cells, of which 98.9% are identified.

What is audited The calibration-matched rows do not decompose the checkpoints as released; they decompose the two models after each is rescaled by its own temperature ($T^* = 2.50$ for the baseline, 0.92 for TDE) fitted on a held-out half of the images. That is the comparison we argue is meaningful, because a proper score punishes miscalibration and these models differ sharply in it. The released pair, scored as it ships, gives a *significant* covariance loss (-0.01355 , interval excluding zero). We report both.

Result. Read raw, TDE loses 0.11651 of quadratic score, of which 0.10296 sits in the transported channel and a small but significant 0.01355 in covariance. That reading does not survive the control. Once temperatures are fitted on held-out images and the audit runs on the other half, the quadratic-score covariance channel is indistinguishable from zero: $\widehat{\Delta R} = -0.00006$ with interval $[-0.00120, +0.00109]$ and randomization $p = .555$. Essentially the entire proper-score difference, -0.18258 of -0.18264 , is transported.

The log-score row of the same calibrated comparison does not repeat the null: it returns $+0.04396$, CI $[+0.04079, +0.04714]$, excluding zero. The two scores therefore disagree on whether any real covariance change survives confidence matching, while agreeing that whichever value is right is far smaller than the calibrated transported shift (-0.18258 under the quadratic score, -0.59157 under the log score). We do not adjudicate: nothing in Section 3.1 privileges one proper score, and Theorem 2 is why this is not a defect to argue away — each Bregman generator defines its own exact estimand, so two scores disagreeing are answering two well-posed questions, and which one an evaluator wants is a declaration. The claim the audit supports is what both scores agree on: TDE’s proper-score improvement over the baseline is overwhelmingly transported *at the calibration-matched, class-pair configuration*, with a covariance remainder at most a small fraction of the total and possibly exactly zero.

That qualifier is load-bearing. Table 1’s five configurations of this one pair give covariance gains of -0.21711 , -0.12619 , -0.01355 , -0.00006 and $+0.04396$, and at two of the five “overwhelmingly transported” is simply false: the raw log-score row has $|\widehat{\Delta R}| = 0.126$ against $|\widehat{\Delta P}| = 0.159$, and the subject-only row has -0.217 dominating $+0.102$. We report all five and argue for one on stated grounds — matched confidence, because a proper score punishes miscalibration; the finest defensible cell, because Proposition 2 makes coarsening non-neutral.

Reading This characterises what TDE does rather than indicting it: its construction subtracts the context-only prediction, so a method operating on the transported channel is what its authors built, and the decomposition quantifies that the intervention is almost purely of that kind. The consequence for evaluation needs care. Mean recall rose 10.2 points; the calibration-matched quadratic covariance gain is -0.00006 with an interval containing zero. Those are different units and we supply no bridge: nothing licenses saying ten points bought 0.00006 of case-level evidence, and it would be equally wrong to say the mean-recall gain *is* transported, since the transported channel on the proper score *fell*. What the audit supports is a conjunction — the headline metric moved ten points, and the proper-score difference decomposes almost entirely into the transported channel — from which a reader may conclude that the metric movement is not evidence about within-cell case-level discrimination, but may not read off a conversion rate.

4.3 Relation prediction with controlled predictors

A released checkpoint cannot tell us whether the decomposition is measuring what it claims, because the ground truth about what the model can see is unavailable. We therefore also compare five predictors on VG150 whose information content is controlled by construction: a frequency table (FREQ) that sees only the class pair, an MLP on class embeddings (MLP-CLASS), and MLPs adding box geometry (MLP-SPATIAL,

Table 1: WAGER audit of released checkpoints on the canonical VG150 PredCls split: MOTIFS-TDE against the MOTIFS-SUM baseline it debiases, both evaluated with the original authors’ code. Raw rows use all identified relations; the calibration-matched rows fit each model’s temperature on a held-out half of the images and audit the disjoint half. CI is image-cluster robust on $\widehat{\Delta R}$.

Regime	Score	Cell ϕ	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$ (95% CI)
raw	quadratic	class pair	−0.11651	−0.10296	−0.01355 [−0.01504, −0.01206]
raw	log	class pair	+0.03277	+0.15897	−0.12619 [−0.13093, −0.12145]
raw	quadratic	subject	−0.11536	+0.10175	−0.21711
calibration-matched	quadratic	class pair	−0.18264	−0.18258	−0.00006 [−0.00120, +0.00109]
calibration-matched	log	class pair	−0.54760	−0.59157	+0.04396 [+0.04079, +0.04714]

MLP-SPATIAL+). Full protocol, per-pair tables and the accuracy figures are in Appendix O.1; the audit covers 227,337 of 229,605 test relations, a coverage of 99.0%, over 6,346 identified class-pair cells.

Two readings matter here. MLP-CLASS improves on *FREQ* by 0.03386 of quadratic score and the decomposition assigns *all* of it to the transported channel: the covariance gain is 0.00000 with a zero-width interval and randomization $p = .724$. That zero is not an empirical finding but an algebraic one — both models are functions of ϕ alone, so their gain vector is constant within each cell and Theorem 1 forces the remainder to vanish — and it is the sharpest available check that the estimator credits nothing to case-level evidence when none exists. Adding box geometry then produces a covariance gain of 0.01439 (CI [0.01373, 0.01504], $p = .002$) that no aggregate score separates out, and against MLP-CLASS the total gain of 0.00885 actually understates it, because the transported channel deteriorates by 0.00554 — the same structure as model B of Appendix A, now on real predictors.

4.4 Real pixels against a geometric proxy

None of the predictors above ever sees an image: they are functions of the class pair and, for MLP-SPATIAL(+), of box coordinates recorded in the annotation. Since box geometry is only a proxy for image content, we also compare a model that reads pixels — subject and object regions cropped from the original image and encoded by a frozen CLIP ViT-B/32 (Radford et al., 2021), concatenated with the class embeddings the class-only model already uses. CLIP is never fine-tuned, so whatever covariance the model gains reflects information its pretraining had already recovered. All three compared predictors are retrained on one fixed subsample of 100,000 training relations so that any gap between them isolates the features rather than the training-set size; the full protocol, tables and figure are in Appendix O.3.

This is the sharpest case in the paper for separating the components. By every aggregate measure the CLIP model is the worst of the three: top-1 accuracy 0.6161 against 0.6467 for the geometry model, and a quadratic score 0.04655 lower. An evaluator reading either number would discard it. The decomposition contradicts that reading on the axis that matters. Against the geometry model the CLIP model loses 0.05296 of transported gain while *gaining* 0.00641 of covariance (95% CI [0.00514, 0.00769], randomization $p = .002$): its entire deficit sits in the transported channel and its within-cell discrimination is significantly better, not worse. Against the class-only baseline, pixels buy 0.01665 of covariance where box geometry buys 0.01023, with well-separated intervals. Real image content carries more case-specific signal than the geometric proxy standing in for it, and a benchmark whose headline metric rewards transported gain will nonetheless rank the pixel model last. Two falsification checks support the reading: post-hoc prior correction registers as almost pure transported movement (+0.02401 against +0.00111), and applied to the CLIP model it halves the aggregate deficit (−0.04655 to −0.02143) while the covariance advantage persists (+0.00752, and +0.00663 when both models are corrected) and survives confidence matching (+0.00467, CI [+0.00314, +0.00621]).

4.5 Other modalities, and sensitivity

Appendix O reports two further studies and the sensitivity analyses. On CIFAR-100-LT, with the benchmark’s coarse superclasses as ϕ and a residual backbone (He et al., 2016), two standard re-weighting schedules have opposite compositions, and the confidence-matched reading reverses what the raw scores suggest about both: the schedule whose aggregate score barely moves has lost within-class discrimination, while the one that improves has bought covariance rather than the prior refit its raw decomposition implies, replicating across seeds and imbalance ratios. A long-tailed split of the 20-Newsgroups corpus (Lang, 1995) runs unmodified and returns interpretable output, but rests on a single trained pair per schedule with no confidence-matched row, so we report it as transfer rather than corroboration. On sensitivity: substituting a floored log score, coarsening ϕ to the subject class, and raising the minimum cell size from 2 to 20 all move the estimate, and the coarsening moves it most, from 0.01439 to 0.02181 — the between-cell covariance term of Appendix E in action. The finest defensible ϕ is therefore the conservative choice, and Section 5 explains why even that rule is weaker than it looks.

5 Discussion and limitations

5.1 What the decomposition does and does not establish

The unit of analysis is a model-pair gain rather than a model, and that is what makes the two failure modes visible. A model may estimate within-group label frequencies better without predicting any individual case better, and be credited as though it had done the second. Conversely it may discriminate cases better while losing enough prior fit to hide the improvement — Section 4.4 contains a case where the aggregate score and the accuracy both point the wrong way. The signed identity $\Delta T = \Delta P + \Delta R$ exposes both without a fitted reference forecaster.

Everything is conditional on the declared ϕ . A positive ΔR means the new model matches its probability changes to the correct case better than the old one *among cases sharing ϕ* ; it proves nothing about causal inference, compositional understanding, or robustness. If an unrecorded shortcut varies inside a cell, its predictive use is counted in ΔR , and Proposition 3 bounds that exposure rather than removing it — at a level that, for our own Visual Genome estimates, is not comfortable (Appendix J). Applications should declare ϕ before seeing results, justify it from how the benchmark was built, and report finer and coarser alternatives, because coarsening adds a between-cell covariance term of either sign (Proposition 2) and only the finest identified partition nets out every regularity expressible in ϕ . Even that rule is weaker than it looks: Appendix L shows two equally fine, fully identified partitions of the same four cases that disagree completely about whether an improvement is case-level or transported, so the choice of ϕ is a modelling decision and not a formality.

Two limits deserve stating flatly. First, ΔR is not invariant to monotone recalibration: it is a covariance between probability movements and labels, so sharpening or softening a fixed model shifts score between the channels while changing no ranking, no top-1 decision and no information about any case. Section 4.1 measures the effect at -0.488 , tens of times any real gain reported here, which is why we audit confidence-matched pairs and why released checkpoints, which routinely differ sharply in calibration, cannot be compared on their raw outputs. A rank-based within-cell contrast would be invariant to this, at the cost of the exact additivity that is the point of the construction; Appendix L says what that trade would involve. Second, a transported gain is not evidence of anything illegitimate. Fitting the deployment label distribution better is real skill whenever that distribution is the one that matters; it is fragile precisely under label shift, where it can be added or removed post hoc by re-weighting (Saerens et al., 2002; Lipton et al., 2018), as Section 4.4 demonstrates and Appendix A exploits. Nor is it purely a matter of prior fit (Proposition 1). The decomposition reports where a gain lives, not whether it is worth having.

5.2 Choosing the grouping variable, and what the estimator cannot do

“Declare ϕ before looking” is advice, not a mechanism. The annotation prior is a property of the dataset, so we recommend that the *benchmark*, not the submitting author, fix it, which removes the incentive to report

only the most favourable candidate; that only moves the discretion, since maintainers often author methods evaluated on their own benchmark. What we would defend is the weaker claim that ϕ be fixed and public *before* a leaderboard opens, by whoever does it, with every pre-declared candidate reported as our sensitivity rows are. Guarantees holding uniformly over a class of groupings are the subject of multicalibration (Hébert-Johnson et al., 2018), and porting that to pairwise attribution is a natural extension. For comparability we suggest a fixed block per audited pair: ϕ and its rationale, the identified coverage, both signed components with the covariance interval, and the confidence-matched and granularity rows.

Exact transport needs at least two cases per cell, so a very fine ϕ buys confound control at the cost of coverage; we report the identified fraction and claim nothing for singletons. The image-clustered interval is asymptotic, and Appendix K justifies it in a regime no heavy-tailed benchmark attains in sample: of 6,346 eligible class-pair cells, 2,286 hold fewer than five cases, though those carry only 2.7% of identified relations. The reported 94.0% empirical coverage against a nominal 95% therefore remains the operative evidence for the interval. The randomization test is finite-group exact only under a sharper within-cell exchangeability null and treats relations rather than images as the randomized units, so it stays a secondary diagnostic.

Finally, the estimator reads probability vectors, not top-1 decisions. That is a feature — it detects useful redistribution below the argmax — but it needs released logits or probabilities, and that is the real obstacle to routine use: the audit of Section 4.2 needed nothing but two checkpoints evaluated with their authors’ code, and what blocks the same workflow elsewhere is disclosure, not compute, since benchmarks overwhelmingly archive ranked predictions. Nothing in Sections 3.1–3.9 is specific to relation prediction, and the long-tailed studies test that in two other modalities. The constraint on ϕ shows its teeth there: the fine class label cannot serve as a cell, since it would leave every cell label-homogeneous and drive ΔR to zero by construction, an artefact that reads exactly like a null result.

6 Conclusion

When two models are compared on a benchmark whose labels are partly determined by a feature both of them observe, the reported gain answers a question nobody asked. It sums a closer fit to the label frequencies within groups of that feature and a closer match between predictions and the individual cases they were made for — different achievements with different consequences, since the first is skill a frequency table also supplies and that a shift in label frequencies removes, and the second is not.

This paper separates them exactly. Scoring each prediction against the label of a different case in the same group leaves the group’s label frequencies intact while breaking the correspondence between prediction and case, so the gain surviving that relabelling is the transported component and the remainder is a within-group covariance gain. The identity holds in the observed sample for any proper score, with no fitted reference model, no held-out fold and no tuning parameter, at $O(NK)$ on cached predictions. The remainder is an order-two U-statistic equal, for every Bregman score, to a within-group covariance between the models’ probability change and the label indicator, and it carries cluster-robust intervals, a randomization test and a limit theorem.

What it buys Pointed at two released MOTIFS checkpoints with their authors’ own code, the decomposition shows a ten-point mean-recall improvement corresponding to a proper-score difference that is overwhelmingly transported. On predictors whose information content we control it returns exactly zero covariance gain when both models are functions of the grouping variable alone, and shows a frozen-CLIP model that loses on every aggregate measure discriminating individual cases significantly better than the model it appears to lose to.

Weaknesses Three are structural. The estimator is not invariant to recalibration, so we compare only confidence-matched pairs and released checkpoints cannot be audited on raw outputs. It credits any signal varying inside a cell, shortcuts included, and our own sensitivity bound is not reassuring: an unrecorded covariate correlated at 0.061 would account for the entire covariance gain we report on Visual Genome. And the estimand is not determined by the granularity of ϕ — two equally fine, fully identified partitions of the

same four cases can disagree completely — so declaring ϕ is a modelling decision that choosing the finest defensible partition does not settle.

Future work A rank-based within-cell contrast would be invariant to the recalibration this construction is sensitive to, at the cost of the exact additivity that is the point here; reporting both is what we would recommend to a benchmark able to choose. Approximate matching would extend the identity to continuous grouping variables, trading exactness for modelling assumptions. And porting multicalibration-style guarantees to pairwise attribution would address the choice of ϕ more convincingly than a declaration rule can. The obstacle to all of it is the one that limits the method today: benchmarks archive ranked predictions, not probabilities.

Broader Impact Statement

This paper supplies an auditing tool, and the way it can go wrong is by being read as a verdict. A small or negative $\widehat{\Delta R}$ says that a reported gain is mostly attributable to fitting label frequencies within cells of one declared grouping variable; it does not say that the gain is spurious, that the authors of the audited system did anything wrong, or that the system is worse. Fitting the deployment label distribution better is real skill whenever that distribution is the one that matters. Three properties in particular limit what an audit licenses, and all three are established in this paper rather than left to the reader: the estimand is conditional on the grouping variable an analyst declares, so a different defensible choice can give a different answer even at equal granularity (Appendix L); the covariance channel credits any signal varying inside a cell, including a shortcut nobody recorded, and our own bound on that exposure is not reassuring (Appendix J); and the estimand is not invariant to recalibration, so an audit of two checkpoints that were not confidence-matched measures partly their calibration and not their evidence. Used against published work without those caveats, the method could support unfair claims about other researchers’ contributions. Used as intended — by a benchmark, on a grouping variable declared before a leaderboard opens, with the confidence-matched row reported alongside — it makes visible a distinction that aggregate metrics currently hide.

Reproducibility Statement

Every experiment uses public data: Visual Genome through the canonical VG150 predicate-classification split, the two publicly released MOTIFS and MOTIFS-TDE checkpoints of Tang et al. (2020) evaluated with their authors’ own code, CIFAR-100 and 20-Newsgroups. No proprietary or restricted data is used. The estimator, its unit tests, every experiment driver, the cached model outputs and a verification script that checks each of the reported numerical values against the committed results files are submitted as anonymized supplementary material; the public repository is withheld here only because it would identify the author, and will be cited on acceptance. Appendix I gives the architectures, schedules, seeds and compute environment.

Use of Generative AI

Generative AI tools were used to support the writing of this manuscript and the development of the accompanying code. The author reviewed and edited the output and takes full responsibility for the content.

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Table 2: A four-point example in a single cell. Model A moves every case to the cell’s label frequencies; model B moves each case towards its own correct label by the same amount. The two models are equally far from q_0 , and only B uses information about the individual case.

case i	label y_i	$q_0(\cdot \mid x_i)$	model A: $q_1(\cdot \mid x_i)$	model B: $q_1(\cdot \mid x_i)$
1	1	(0.5, 0.5)	(0.75, 0.25)	(0.75, 0.25)
2	1	(0.5, 0.5)	(0.75, 0.25)	(0.75, 0.25)
3	1	(0.5, 0.5)	(0.75, 0.25)	(0.75, 0.25)
4	2	(0.5, 0.5)	(0.75, 0.25)	(0.25, 0.75)

A A worked example

Four data points are enough to show the whole construction, and to exhibit in arithmetic the main properties Section 3 proves in general. Take one cell containing four test cases and two possible labels. Three of the cases carry label 1 and one carries label 2, so the cell’s label frequencies are $(3/4, 1/4)$. The old model q_0 is uninformative within the cell: it predicts $(0.5, 0.5)$ for all four cases. We compare two candidate replacements, shown in Table 2.

Score both models with the quadratic score $S(q, y) = 2q_y - \|q\|_2^2$, and for each case tabulate the score *difference* at each of the two possible labels, not only at the observed one. Writing $h_i(y) = S(q_1(\cdot \mid x_i), y) - S(q_0(\cdot \mid x_i), y)$, all the arithmetic is in eighths:

$$h_i(y) = \begin{cases} +3/8 & \text{if } y \text{ is the label the new model moved towards,} \\ -5/8 & \text{otherwise.} \end{cases} \quad (25)$$

This vector, evaluable at labels that were never observed for case i , is the only object the method needs beyond the labels themselves.

Model A: a pure prior refit. Every case gets the same prediction, so $h_i(1) = +3/8$ and $h_i(2) = -5/8$ throughout, and the observed mean gain is $\Delta T = \frac{1}{4}[3 \cdot \frac{3}{8} - \frac{5}{8}] = \frac{1}{8}$. Now transport: replace each case’s label in turn by each of the other three and average. Cases 1–3 are each scored against $\{1, 1, 2\}$, giving $\frac{1}{3}(\frac{3}{8} + \frac{3}{8} - \frac{5}{8}) = \frac{1}{24}$; case 4 is scored against $\{1, 1, 1\}$, giving $+\frac{3}{8}$. So $\Delta P = \frac{1}{4}[3 \cdot \frac{1}{24} + \frac{3}{8}] = \frac{1}{8}$ and $\Delta R = 0$ — exactly, in this sample, not on average over hypothetical resamples. That is the general fact: a prediction change constant within a cell contributes nothing to ΔR (Theorem 1). Note also that $\Delta P = 1/8$ is the largest transported gain any within-cell-constant prediction can earn here, since a proper score is maximised in expectation at the true frequencies $(3/4, 1/4)$ — which is what A predicts.

Model B: a case-specific improvement. Cases 1–3 are unchanged, but case 4 now moves towards label 2, so $h_4(1) = -5/8$ and $h_4(2) = +3/8$. Every case is scored at $+3/8$ on its own label, so $\Delta T = 3/8$. Under transport cases 1–3 are unaffected, but case 4 is scored against three label-1 cases at $-5/8$ apiece, so $\Delta P = \frac{1}{4}[3 \cdot \frac{1}{24} - \frac{5}{8}] = -\frac{1}{8}$ and $\Delta R = \frac{1}{2}$.

Two things follow. First, B’s aggregate gain of $3/8$ *understates* its case-level improvement of $1/2$, and not because B fits the cell frequencies worse — it fits them better, moving the cell-mean prediction from 0.5 to 0.625 against a truth of 0.75, an improvement of exactly 0.09375. What offsets it is dispersion: B’s forecasts vary inside the cell where the old model’s did not, and by Proposition 1 the transported channel charges that variation at the same 0.09375, so the two cancel and the in-sample transported gain is zero. The $-1/8$ the estimator reports is what remains once each case is scored against the *other* cases’ labels only, removing the $\frac{1}{n_c}\widehat{\Delta R} = 1/8$ the in-sample plug-in had credited to the transported channel. Second, an evaluator reading only the total would credit B with three times A’s improvement, whereas the two improve on disjoint grounds: A earns $1/8$ purely by moving to the cell frequencies and gains no covariance, while B’s $1/2$ of covariance is partly concealed by a transported term that turns negative. This is the failure mode that motivates the paper, in miniature — aggregate scores are not merely imprecise about composition, they can order two models by the wrong criterion.

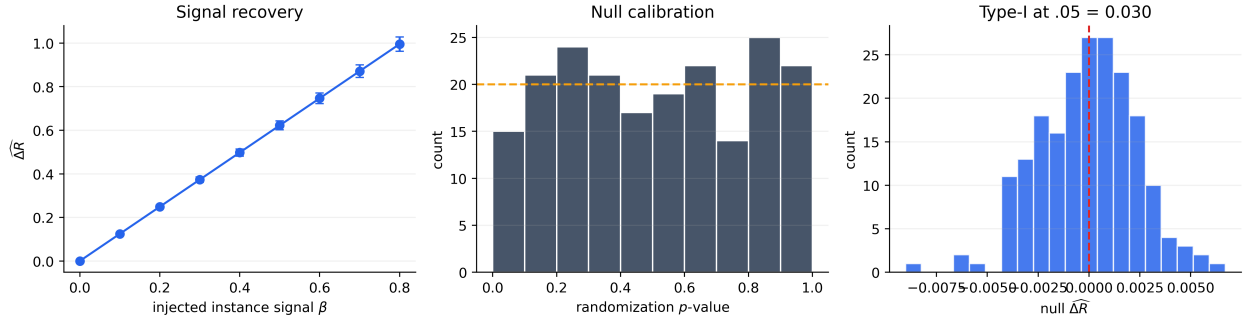


Figure 2: **Controlled validation.** WAGER recovers injected within-cell signal monotonically (mean $\pm$ one standard deviation), produces approximately uniform null randomization p -values, and controls type-I error in 200 null runs.

The covariance form, and why leaving one out matters. Theorem 1 says ΔR is a within-cell covariance between the probability change and the label indicator. Computing it directly for model B gives $2 \sum_y \text{Cov}(\Delta q_y, \mathbf{1}\{Y = y\}) = 3/8$, not the $1/2$ transport returned. Neither is wrong: the covariance uses the cell’s *own* four labels to form its reference, so each case contributes to the reference it is compared against, and that in-sample plug-in is attenuated by exactly $(n_c - 1)/n_c = 3/4$ — indeed $\frac{3}{4} \cdot \frac{1}{2} = \frac{3}{8}$. Transport, which scores each case only against the *other* cases’ labels, removes the attenuation with no held-out data. Proposition 4 shows this holds for every cell with $n_c \geq 2$ and every grouping variable, which is why the method needs no sample splitting. The example is reproduced as a unit test in the accompanying software.

One property the example cannot show, because both candidates happen to be equally confident, is the estimator’s sensitivity to confidence itself. Multiply every prediction’s odds by a constant and ΔR moves, though no case has been reordered and no information added; Section 4.1 measures that at -0.488 in a design where the two models carry identical case-level information. Every result in this paper is therefore reported for confidence-matched pairs.

A decision the split changes. At the benchmark’s own label frequencies both candidates improve and B improves three times as much, so an evaluator reading the score picks B, rightly. Suppose deployment sees label frequencies in this cell of $(\frac{1}{4}, \frac{3}{4})$ instead of the observed $(\frac{3}{4}, \frac{1}{4})$ — a label shift, the situation post-hoc prior correction addresses (Saerens et al., 2002; Lipton et al., 2018). Reweighting each case by $p'(y_i)/p(y_i)$ and rescore sends A from $+\frac{1}{8}$ to $-\frac{3}{8}$ while leaving B at $+\frac{3}{8}$: A becomes *worse* than the model it was meant to replace, on a change that leaves every prediction exactly as it was. The decomposition says in advance which is exposed and the aggregate score does not, because A’s whole gain sits in the transported channel — the one a change in label frequencies acts on — and B’s in the covariance channel, which such a change leaves alone. That is the operational content of the split: $\widehat{\Delta P}$ is the part of a reported improvement contingent on the evaluation set’s label frequencies matching deployment, and $\widehat{\Delta R}$ is the part that is not. Section 4.4 exhibits the same structure on real models.

B Bregman-score identity

Eq. (10) gives the log-score check of Section 4.2 the same exact covariance meaning that Eq. (7) gives the quadratic score: within-group covariance under the log score is the within-cell covariance between the log-likelihood-ratio contrast of the two models and the observed label indicator, not merely "a sensitivity analysis." Any other Bregman score — the spherical or power-family scores catalogued by Gneiting & Raftery (2007), for instance — inherits an identity of the same form through its own ψ , so Theorem 2 fixes the scope of the estimator’s scoring-rule dependence once, rather than leaving each alternative score to be checked separately. The implemented log score floors each probability at $\epsilon = 10^{-6}$ to avoid $-\infty$ on an unseen label; Eq. (10) is exact for the unfloored $\log q_y$, and the floored version used throughout is a sensitivity check on

that exact identity rather than a literal instance of it, negligibly different whenever no relevant probability is within ϵ of zero.

C Finite-sample identity

D Randomization test

As a complementary finite-group diagnostic, WAGER applies an independently sampled cyclic shift to labels inside every cell. Each shift preserves cell size and label histogram. Recomputing $\widehat{\Delta R}$ under B shifts gives

$$p = \frac{1 + \sum_{b=1}^B \mathbb{1}\{\widehat{\Delta R}^{(b)} \geq \widehat{\Delta R}\}}{B + 1}. \quad (26)$$

This is exact under the sharp null that labels are exchangeable relative to the frozen model outputs within each cell. Because SGG relations share images, our primary real-data uncertainty statement is the image-cluster-robust interval; the randomization test is reported as a secondary relation-level diagnostic.

E Coarsening and unrecorded covariates

Appendix O.4 reports that replacing the class-pair cell with a coarser subject-only cell changes the estimated covariance gain. The following result gives that sensitivity an exact population-level mechanism rather than treating it as an empirical curiosity.

The first term is the coarse cell’s share of the fine-grained covariance gain; the second is a between-cell covariance, over the finer cells nested inside $\bar{c}$, between each fine cell’s typical gain at label y and its label- y frequency. It is not sign definite, so coarsening ϕ can inflate or deflate the credited covariance gain, and it vanishes exactly when either quantity is constant within each coarse cell. This is why the finest defensible ϕ is the conservative default: coarsening adds cross-cell prior structure into ΔR , while the finer partition’s gain is preserved on average through the first term. The proof is a per-label application of the law of total covariance (Appendix F).

E.1 Sensitivity to an unrecorded confounder

Read in one direction, Proposition 2 says what folding a covariate into ϕ does. Read in the other, it quantifies what happens when a relevant covariate is *never* declared, which is the estimator’s main scope limit. Let Z be unrecorded and let (ϕ, Z) be the finer, fully adjusted cell. Applying Proposition 2 with $\tilde{\phi} = \phi$ gives, for every declared cell c ,

$$\Delta R_c(\phi) = \mathbb{E}[\Delta R_{(\phi, Z)} \mid \phi = c] + \sum_{y=1}^K \text{Cov}(\mathbb{E}[H_X(y) \mid \phi, Z], \Pr(Y = y \mid \phi, Z) \mid \phi = c), \quad (27)$$

whose first term is what an analyst who could observe Z would report and whose second is the bias of using ϕ alone. The identity is exact but needs Z ’s joint law with (H, Y) . The following bound replaces that with a judgement about how strongly an unrecorded variable could plausibly act, in the spirit of Rosenbaum (2002).

Proposition 3 states that bound and Corollary 2 its data-free form; both are proved in Appendix F. Because $\bar{\rho}$ is a correlation rather than a domain-specific odds ratio, it can be elicited the same way across benchmarks. Appendix J evaluates both forms on our own estimates, and the sharp one is the less comfortable: an unrecorded covariate correlated at 0.061 would account for the entire covariance gain reported in Appendix O.2.

F Proofs

F.1 Proof of Theorem 1

Condition throughout on $C = c$. Expanding over labels gives

$$\Delta T_c = \sum_y \mathbb{E}[H_X(y) \mathbb{1}\{Y = y\}], \quad \Delta P_c = \sum_y \mathbb{E}[H_X(y)] \Pr(Y = y).$$

Subtracting yields the covariance sum in Eq. (6); the first identity is the definition of ΔR_c . Under the quadratic score,

$$H_X(y) = 2\Delta q_y(X) - \{\|q_1(\cdot | X)\|_2^2 - \|q_0(\cdot | X)\|_2^2\}.$$

The bracketed term is independent of the label index. Its contribution summed over y is $\text{Cov}(B(X), \sum_y \mathbb{1}\{Y = y\}) = \text{Cov}(B(X), 1) = 0$. The remaining terms give Eq. (7). If $H_X(y) = H_c(y)$ almost surely, every covariance is zero. $\square$

F.2 Proof of Proposition 1

Condition on $C = c$ and let $Y' \sim p(c)$ be independent of X . For the quadratic score $S_B(q, y) = 2q_y - \|q\|^2$,

$$\mathbb{E}[S_B(q_m, Y') | c] = 2\langle p(c), \bar{q}_m(c) \rangle - \mathbb{E}[\|q_m\|^2 | c]. \quad (28)$$

Decompose the second moment as $\mathbb{E}[\|q_m\|^2 | c] = \|\bar{q}_m(c)\|^2 + \text{tr Var}(q_m | c)$, and complete the square using $2\langle p, \bar{q} \rangle - \|\bar{q}\|^2 = \|p\|^2 - \|p - \bar{q}\|^2$:

$$\mathbb{E}[S_B(q_m, Y') | c] = \|p(c)\|^2 - \|p(c) - \bar{q}_m(c)\|^2 - \text{tr Var}(q_m | c). \quad (29)$$

Subtracting the $m = 0$ expression from the $m = 1$ expression, the $\|p(c)\|^2$ terms cancel and Eq. (8) follows, since $\Delta P_c = \mathbb{E}[S_B(q_1, Y') | c] - \mathbb{E}[S_B(q_0, Y') | c]$ by definition. $\square$

F.3 Proof of Theorem 2

Condition on $C = c$. The proof of Theorem 1 above establishes $\Delta R_c = \sum_y \text{Cov}(H_X(y), \mathbb{1}\{Y = y\} | C = c)$ for *any* score contrast H , not only the quadratic one; only the identification of $H_x(y)$ with a specific functional form is score-dependent. For the Bregman score of the Bregman score $S_\psi(q, y) = \psi(q) + \langle \nabla \psi(q), e_y - q \rangle$,

$$H_x(y) = \underbrace{[\psi(q_1(x)) - \psi(q_0(x))] - [\langle \nabla \psi(q_1(x)), q_1(x) \rangle - \langle \nabla \psi(q_0(x)), q_0(x) \rangle]}_{=: B(x)} + \Delta g_y(x),$$

writing $q_i(x)$ for $q_i(\cdot | x)$, where $B(x)$ collects every term not indexed by y and $\Delta g_y(x) = \partial_y \psi(q_1(x)) - \partial_y \psi(q_0(x))$ is the y -indexed remainder, exactly as in the proof of Theorem 1 for the quadratic case. Substituting,

$$\begin{aligned} \sum_y \text{Cov}(H_X(y), \mathbb{1}\{Y = y\} | c) &= \text{Cov}\left(B(X), \sum_y \mathbb{1}\{Y = y\} \mid c\right) \\ &\quad + \sum_y \text{Cov}(\Delta g_y(X), \mathbb{1}\{Y = y\} | c). \end{aligned}$$

The first term is $\text{Cov}(B(X), 1 | c) = 0$ since $\sum_y \mathbb{1}\{Y = y\} = 1$ almost surely. The second term is Eq. (9). $\square$

F.4 Proof of Theorem 3

Summing the kernel in Eq. (19) over unordered pairs, every observed term $H_i(y_i)$ occurs in $n_c - 1$ pairs with coefficient $1/2$, while the two crossed terms together enumerate every ordered pair $H_i(y_j)$, $i \neq j$, with coefficient $1/2$. Since $\binom{n_c}{2} = n_c(n_c - 1)/2$,

$$\widehat{\Delta R}_c = \frac{1}{n_c} \sum_i H_i(y_i) - \frac{1}{n_c(n_c - 1)} \sum_{i \neq j} H_i(y_j) = \widehat{\Delta T}_c - \widehat{\Delta P}_c.$$

This is deterministic. When the cell's examples are i.i.d. draws from the cell population, the average of the symmetric order-two kernel is the standard unbiased U-statistic for its population expectation Hoeffding (1948); independence is used, not merely exchangeability, because ΔP_c is defined through an independent label coupling, so $\mathbb{E}[H_1(Y_2)] = \mathbb{E}[H_X(Y')]$ requires the two draws to be independent. $\square$

F.5 Proof of Proposition 2

Fix y and condition throughout on $\bar{\phi} = \bar{c}$. The law of total covariance applied to the pair $(H_X(y), \mathbb{1}\{Y = y\})$ with the finer conditioning variable C nested inside $\bar{c}$ gives

$$\begin{aligned} \text{Cov}(H_X(y), \mathbb{1}\{Y = y\} \mid \bar{\phi} = \bar{c}) &= \mathbb{E}[\text{Cov}(H_X(y), \mathbb{1}\{Y = y\} \mid C) \mid \bar{\phi} = \bar{c}] \\ &\quad + \text{Cov}(\mathbb{E}[H_X(y) \mid C], \Pr(Y = y \mid C) \mid \bar{\phi} = \bar{c}), \end{aligned}$$

using $\mathbb{E}[\mathbb{1}\{Y = y\} \mid C] = \Pr(Y = y \mid C)$. Summing over $y = 1, \dots, K$ and applying the population identity of Theorem 1 to the left side (at $\bar{\phi} = \bar{c}$) and to the first term on the right side (at $C = c$, then averaged over c within $\bar{c}$) yields Eq. (11). $\square$

F.6 Proof of Proposition 4

Fix a cell c with $n_c \geq 2$ and $i \in c$. By definition of the label count, $\sum_{y=1}^K m_{cy} H_i(y) = \sum_{j \in c} H_i(y_j) = H_i(y_i) + \sum_{j \neq i} H_i(y_j)$, and Eq. (21) gives $\sum_{j \neq i} H_i(y_j) = (n_c - 1)\hat{P}_i$. Hence

$$\tilde{P}_i = \frac{1}{n_c} \sum_y m_{cy} H_i(y) = \frac{H_i(y_i) + (n_c - 1)\hat{P}_i}{n_c}.$$

Substituting $H_i(y_i) = \hat{P}_i + \hat{R}_i$ gives $\tilde{P}_i = \hat{P}_i + n_c^{-1}\hat{R}_i$, and $\tilde{R}_i = H_i(y_i) - \tilde{P}_i = \hat{R}_i - n_c^{-1}\hat{R}_i = (n_c - 1)n_c^{-1}\hat{R}_i$. Averaging $\tilde{R}_i$ within cell c and reweighting by n_c across cells, as in the dataset-level definition of $\widehat{\Delta Z}$, gives $\widehat{\Delta R} = N_*^{-1} \sum_c n_c \cdot \frac{n_c - 1}{n_c} \widehat{\Delta R}_c = N_*^{-1} \sum_c (n_c - 1) \widehat{\Delta R}_c$. $\square$

F.7 Proof of Proposition 3

Fix $\phi = c$ throughout and write $a_y = \mathbb{E}[H_X(y) \mid \phi, Z]$, $b_y = \Pr(Y = y \mid \phi, Z)$, both random through their dependence on (ϕ, Z) . Let $U = (a_y - \mathbb{E}[a_y \mid c])_{y=1}^K$ and $V = (b_y - \mathbb{E}[b_y \mid c])_{y=1}^K$ be the corresponding mean-zero $\mathbb{R}^K$ -valued random vectors conditional on $\phi = c$. The map $(U, V) \mapsto \mathbb{E}[\langle U, V \rangle \mid c] = \sum_y \text{Cov}(a_y, b_y \mid c)$ is an inner product on the space of mean-square-integrable $\mathbb{R}^K$ -valued random variables conditional on $\phi = c$, so by the Cauchy-Schwarz inequality applied to that inner product,

$$\left| \sum_{y=1}^K \text{Cov}(a_y, b_y \mid c) \right| \leq \sqrt{\mathbb{E}[\|U\|^2 \mid c]} \sqrt{\mathbb{E}[\|V\|^2 \mid c]} = \sqrt{\sum_y \text{Var}(a_y \mid c)} \sqrt{\sum_y \text{Var}(b_y \mid c)}.$$

By the definition of ρ , the left side equals $|\rho| \sqrt{\sum_y \text{Var}(a_y \mid c)} \sqrt{\sum_y \text{Var}(b_y \mid c)}$ exactly, so this step alone is not yet a bound; two further steps supply one, at two different levels of conservatism. First, $a_y = \mathbb{E}[H_X(y) \mid \phi, Z]$ is itself a conditional expectation of $H_X(y)$ given more information than $\phi = c$ alone, so the law of total variance gives $\text{Var}(H_X(y) \mid c) = \mathbb{E}[\text{Var}(H_X(y) \mid \phi, Z) \mid c] + \text{Var}(a_y \mid c) \geq \text{Var}(a_y \mid c)$; likewise $b_y = \Pr(Y = y \mid \phi, Z)$ satisfies $\text{Var}(b_y \mid c) \leq \text{Var}(\mathbb{1}\{Y = y\} \mid c) = p_y(c)(1 - p_y(c))$ by the same argument applied to $\mathbb{1}\{Y = y\}$ in place of $H_X(y)$. Both bounds hold termwise, so summing over y and applying the (monotone) square root to each sum separately gives the observable, per-cell bound

$$\sqrt{\sum_y \text{Var}(a_y \mid c)} \sqrt{\sum_y \text{Var}(b_y \mid c)} \leq \sqrt{\sum_y \text{Var}(H_X(y) \mid c)} \sqrt{\sum_y p_y(c)(1 - p_y(c))},$$

the second inequality of Eq. (13); this is a bound on a product of two sums, and does not simplify to a sum of per-label square roots, since Cauchy-Schwarz runs the other way for that rearrangement. Second,

for Corollary 2, $a_y \in [-M, M]$ as a conditional expectation of a $[-M, M]$ -valued variable, so Popoviciu's inequality gives $\text{Var}(a_y | c) \leq M^2$ for every y and hence $\sum_y \text{Var}(a_y | c) \leq KM^2$. For the b_y the same argument would give $\text{Var}(b_y | c) \leq 1/4$ and a total of $K/4$, but that treats the b_y as K unconstrained quantities in $[0, 1]$ when in fact $\sum_y b_y = 1$ almost surely. Using that constraint, $\sum_y b_y^2 \leq (\sum_y b_y)^2 = 1$ pointwise, while $\sum_y (\mathbb{E}b_y)^2 \geq K^{-1}(\sum_y \mathbb{E}b_y)^2 = 1/K$ by Cauchy–Schwarz, so

$$\sum_{y=1}^K \text{Var}(b_y | c) = \mathbb{E} \left[\sum_y b_y^2 \right] - \sum_y (\mathbb{E}b_y)^2 \leq 1 - \frac{1}{K},$$

which is Eq. (14) and is strictly smaller than $K/4$ for every $K > 2$. Therefore

$$\left| \sum_{y=1}^K \text{Cov}(a_y, b_y | c) \right| \leq |\rho| \sqrt{KM^2} \sqrt{1 - 1/K} = |\rho| M \sqrt{K - 1}.$$

By Eq. (11), the left side of each display is exactly $|\Delta R_c(\phi) - \mathbb{E}[\Delta R_{(\phi, Z)} | \phi = c]|$, giving Eq. (13) and Eq. (15). $\square$

G Influence function: estimand and derivation

The estimand under singleton exclusion. Cells with $n_c = 1$ identify no within-cell counterfactual and are excluded, so the dataset-level statistic targets the *identified-subpopulation* quantity: writing π_c for the cell probabilities, the target of $\widehat{\Delta R}$ is the π -weighted mean $\theta = \sum_c \pi_c \Delta R_c / \sum_c \pi_c$ taken over cells that appear with at least two examples. At finite N the identified set is random; we treat the interval as conditional on identification and report the identified fraction alongside every estimate (99.0% of relations on VG150, 100% on CIFAR-100-LT), so the gap between the conditional and full-population targets is bounded by the reported coverage.

Derivation of Eq. (23). Fix a cell c and let $\theta_c = \Delta R_c$ with $g_c(z) = \mathbb{E}[A(z, Z') | Z' \in c]$ the kernel projection. The classical Hájek projection of the order-two U-statistic $\widehat{\Delta R}_c$ is

$$\widehat{\Delta R}_c = \theta_c + \frac{2}{n_c} \sum_{i \in c} (g_c(Z_i) - \theta_c) + O_p(n_c^{-1}).$$

The dataset statistic weights cells by their example counts, $\widehat{\Delta R} = N_*^{-1} \sum_c n_c \widehat{\Delta R}_c$, and cell membership is itself random (multinomial over cells), so an example i contributes both through the within-cell projection and through the between-cell composition term $\theta_{C_i} - \theta$. Combining the two sources,

$$\widehat{\Delta R} - \theta = \frac{1}{N_*} \sum_i \left[2(g_{C_i}(Z_i) - \theta_{C_i}) + (\theta_{C_i} - \theta) \right] + o_p(N_*^{-1/2}),$$

where the dataset-level remainder requires one condition beyond the per-cell expansion, because C per-cell remainders do not aggregate for free. Writing $C = \#\{c : n_c \geq 2\}$ for the number of eligible cells, the weighted sum of the per-cell degenerate terms is $N_*^{-1} \sum_c n_c \cdot O_p(n_c^{-1}) = O_p(C/N_*)$, which is $o_p(N_*^{-1/2})$ provided

$$C = o(\sqrt{N_*}). \quad (30)$$

This is what makes the second-order rate $O_p(n_c^{-1})$ rather than $o_p(n_c^{-1/2})$ worth stating: the sharper per-cell rate is what allows a condition on the cell *count* alone, with no further requirement on individual cell sizes. On VG150, $C = 6,346$ against $N_* = 227,337$, so $C/\sqrt{N_*} \approx 13.3$ — the condition is a statement about the limit, and at these finite sample sizes it is the coverage simulation of Section 4.1, not Eq. (30), that supports the reported intervals. Estimating $g_{C_i}(Z_i)$ by the pair-kernel mean $\bar{A}_i$, θ_c by $\widehat{\Delta R}_c$, and θ by $\widehat{\Delta R}$ gives the plug-in influence contribution $\widehat{\psi}_i = 2(\bar{A}_i - \widehat{\Delta R}_c) + (\widehat{\Delta R}_c - \widehat{\Delta R}) = 2\bar{A}_i - \widehat{\Delta R}_c - \widehat{\Delta R}$, which is Eq. (23) and has exactly zero sample mean by construction. Its empirical 95% coverage in a VG-like regime (image-clustered examples, heavy-tailed cell sizes with a large mass of near-minimum cells) is validated in Section 4.1.

Proof of Theorem 4. The linearization above gives $\widehat{\Delta R} - \theta = N_*^{-1} \sum_i \psi_i + o_p(N_*^{-1/2})$, under Eq. (30), for the population influence contribution $\psi_i = 2(g_{C_i}(Z_i) - \theta_{C_i}) + (\theta_{C_i} - \theta)$, which has zero mean by construction and satisfies $|\psi_i| \leq C(M)$ for an absolute constant $C(M)$ depending only on the score bound M in condition (i), since g_{C_i} , θ_{C_i} , and θ are themselves averages of the bounded kernel $A_{ij} \in [-2M, 2M]$. Because clusters are mutually independent (condition (ii)), $T_g = \sum_{i \in g} \psi_i$ for $g = 1, \dots, G$ are independent, mean-zero random variables with $|T_g| \leq |g| C(M)$. Condition (ii)’s finite third moment for cluster sizes and $\max_g |g| = o(\sqrt{G})$ give $\sum_g E|T_g|^3 \leq C(M)^3 \sum_g \mathbb{E}|g|^3 = O(G)$ while $(\sum_g \text{Var}(T_g))^{3/2} = \Theta(G^{3/2})$ under condition (iv), so Lyapunov’s ratio $\sum_g E|T_g|^3 / (\sum_g \text{Var}(T_g))^{3/2} \rightarrow 0$; the Lyapunov condition implies the Lindeberg condition, and the Lindeberg–Feller central limit theorem for triangular arrays of independent summands (Serfling, 1980; van der Vaart, 1998) gives $(\sum_g T_g) / \sqrt{\sum_g \text{Var}(T_g)} \xrightarrow{d} \mathcal{N}(0, 1)$. Condition (iii) makes $N_* / \sum_g \text{Var}(T_g) \rightarrow 1/\sigma^2$ in probability by definition of σ^2 , and the $o_p(N_*^{-1/2})$ remainder is asymptotically negligible after multiplying by $\sqrt{N_*}$, so $\sqrt{N_*}(\widehat{\Delta R} - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$ by Slutsky’s theorem. For the variance estimator, $\widehat{\Delta R} \rightarrow_p \theta$ by the weak law of large numbers across the diverging number of identified examples, and $\widehat{\Delta R}_c \rightarrow_p \Delta R_c$ for each eligible cell by the weak law of large numbers applied within that cell – this is where condition (iv) is used, and it is not implied by anything aggregate: (iv) guarantees $n_c \rightarrow \infty$ almost surely for every cell in the fixed, finite support of ϕ , which is what a within-cell WLLN requires, whereas a condition on the total identified count bounds no individual cell’s size. So $\widehat{\psi}_i \rightarrow_p \psi_i$ pointwise, and $\widehat{\sigma}^2$ – a sum of squares of cluster-aggregated $\widehat{\psi}_i$, normalized by N_* and G – is a continuous functional of these consistent plug-ins, hence $\widehat{\sigma}^2 \rightarrow_p \sigma^2$ by the continuous mapping theorem. $\square$

Efficient computation. The implementation does not instantiate all pairs. In addition to Eq. (21), let $G_c(y) = \sum_{j \in c} H_j(y)$ and $O_c = \sum_{j \in c} H_j(y_j)$. The pair-kernel mean for observation i is

$$\bar{A}_i = \frac{1}{2} \left[H_i(y_i) + \frac{O_c - H_i(y_i)}{n_c - 1} - P_i - \frac{G_c(y_i) - H_i(y_i)}{n_c - 1} \right].$$

Therefore both the point estimate and Hájek projection require only label counts, column sums, and matrix–vector products inside each cell. For image clusters g , the implemented variance is

$$\widehat{\text{Var}}(\widehat{\Delta R}) = \frac{G}{G-1} \frac{1}{N_*^2} \sum_{g=1}^G \left(\sum_{i \in g} \widehat{\psi}_i \right)^2,$$

where G is the number of images and $\widehat{\psi}_i$ is Eq. (23). We use a normal critical value; with 27,955 image clusters, small- G corrections are immaterial.

H Randomization algorithm

For each cell, sort examples once and index them $0, \dots, n_c - 1$. Randomization b draws $o_c^{(b)}$ uniformly from this index set and maps label position r to $(r + o_c^{(b)}) \bmod n_c$. Independent offsets across cells define an element of the product of cyclic groups. The cell label histogram is unchanged, so $\sum_y m_{cy} H_i(y)$ in Eq. (21) is precomputed; only the assigned label lookup changes. One randomization is $O(N)$ after the $O(NK)$ setup. The plus-one Monte Carlo correction includes the observed assignment and prevents zero p -values.

I Implementation and reproducibility

The NumPy implementation is in `wager/antisymmetric.py`. The primary routine `decompose_gain` validates probability matrices, forms score contrasts, excludes singleton cells, computes all three components, verifies the identity numerically, and returns per-instance transported/covariance values plus intervals. The Visual Genome driver is `experiments/run_sgg_wager.py`; controlled validation and sensitivity checks are in `experiments/antisymmetric_simulation.py` and `experiments/antisymmetric_ablations.py`.

Algorithm 1 WAGER gain decomposition**Require:** frozen q_1, q_0 ; labels y ; cells ϕ ; proper score S **Ensure:** $\widehat{\Delta T}, \widehat{\Delta P}, \widehat{\Delta R}$ and uncertainty

```

1:  $H_i(k) \leftarrow S(q_{1i}, k) - S(q_{0i}, k)$  for all  $i, k$ 
2: for each cell  $c$  with  $n_c \geq 2$  do
3:   count labels  $m_{c1}, \dots, m_{cK}$ 
4:   for each  $i \in c$  do
5:      $P_i \leftarrow (\sum_k m_{ck} H_i(k) - H_i(y_i)) / (n_c - 1)$ 
6:      $R_i \leftarrow H_i(y_i) - P_i$ 
7:   end for
8: end for
9:  $\widehat{\Delta T} \leftarrow \text{mean } H_i(y_i)$ ;  $\widehat{\Delta P} \leftarrow \text{mean } P_i$ ;  $\widehat{\Delta R} \leftarrow \text{mean } R_i$ 
10: compute image-clustered Hájek interval and cell-shift randomization  $p$ 

```

All random seeds are fixed. Unit tests cover the decomposition identity, quadratic covariance identity, cell-constant null, singleton reporting, cluster inference, randomization power, and log-score path, plus direct numerical checks of Propositions 4 and 2 (the latter against an explicit discrete population with exact expectations, independent of the estimator’s own code path), Corollary 1’s log-score covariance identity, Corollary 3’s equivalence to the (unclustered) Diebold–Mariano statistic at a trivial grouping variable, and Proposition 3’s bound ordering (exact bias $\leq$ tight, per-cell bound $\leq$ crude, data-free bound) on a second explicit discrete population with a known omitted confounder. The covariance-identity cross-check quoted in Section O.5 is reproduced by `experiments/cifar_covariance_check.py`; the calibration-matched protocol and the prior-matching experiment are `experiments/cifar_recalibration_control.py` and `experiments/vg_prior_consequence.py`. Table 3’s free-bound rows are reproduced by `experiments/sensitivity_bound_check.py` and its sharp rows, including the robustness value $\rho^\dagger$ of Eq. (31), by `experiments/sensitivity_analysis.py`; the evaluation of Corollary 2 is `experiments/sensitivity_bound_check.py`, which recomputes the crude bound directly from the committed `antisymmetric_ablations.json` rather than re-running the decomposition; `experiments/sensitivity_analysis.py` computes Proposition 3’s tighter, per-cell form from per-instance arrays (Section I.1).

Full-data VG predictors (Sections O.1–O.4). FREQ uses add-one smoothing over the 50 predicates within each training class pair, backing off to the subject-marginal and then the global predicate histogram for pairs unseen in training. The MLPs consume fixed random class embeddings (32 dimensions per class, seed 0, concatenated for subject and object) on standardized features and train with AdamW (learning rate 10^{-3} , weight decay 10^{-5} , batch size 4096): MLP-CLASS with hidden sizes (256, 128) for 12 epochs, MLP-SPATIAL with (256, 128) for 15 epochs, and MLP-SPATIAL+ with (512, 256, 128) for 20 epochs – the “+” variant differs in both width and schedule, as noted in Section 4.3.

Data availability. The repository ships the estimator, every driver script, and the cached results files that every printed number traces to (`results/*.json`). The cached per-relation probability arrays that the drivers consume (`sgg_dists.npz`, `vg_visual_models.npz`, `cifar_lt_results.npz`) are regenerable from the committed training scripts with fixed seeds and are archived alongside the code release, so exact reproduction does not require retraining.

I.1 Compute environment for Sections O.3–O.5

Model training and feature extraction for both new experiments ran on a single NVIDIA T4 (15 GB) GPU. The WAGER decomposition itself is pure NumPy and requires no accelerator: it consumes the resulting cached probability arrays, as in Section 4.3. All Python code runs under a pinned environment (Python 3.13).

Matched-subsample models (Section 4.4). The three compared predictors share a fixed, seeded (seed = 0) subsample of 100,000 of the 532,987 training relations, drawn without replacement uniformly at the relation level (not the image level); the full 229,605-relation test split is always used for evaluation. All three use the identical head and optimizer – AdamW, learning rate 10^{-3} , weight decay 10^{-5} , 15 epochs, batch size 4096, hidden sizes (256, 128), on standardized features – so they differ only in their input features.

Subject/object crops are taken from the annotated boxes and CLIP-preprocessed (bicubic resize to 224×224 with CLIP’s published normalization constants); a crop with no valid overlap with the image falls back to a blank crop rather than being dropped, so every relation contributes a feature row. CLIP runs in half precision with a batch size of 512 crops and is never fine-tuned. Two implementation details are worth recording. First, the pretrained OpenAI weights were trained with the QuickGELU activation, so the encoder must be instantiated as ViT-B-32-quickgelu; loading the same weights under the plain ViT-B-32 configuration silently substitutes a standard GELU and degrades every embedding. Second, because an annotated object box participates in several relations, crops are deduplicated by (image, box) before encoding and gathered afterwards: the 659,210 crops implied by the audited relations collapse to 422,143 unique ones, removing 36.0% of the encoder work without changing any feature. Only the images referenced by those relations are retrieved, individually rather than through the two bulk archives; of the 71,990 required, three were unavailable upstream and fall back to a blank crop.

ResNet-32 triple (Section O.5). CIFAR-100-LT uses the standard exponential-profile construction of Cui et al. (2019) at imbalance ratio 100: class c (zero-indexed, $0, \dots, 99$) keeps $n_c = \lfloor 500 \cdot \mu^c \rfloor$ training images, where $\mu = 100^{-1/99}$, so class 0 keeps all 500 images and class 99 keeps 5; the balanced 10,000-image test split is untouched. All three models are a from-scratch, randomly initialized CIFAR ResNet-32 (three stages of five basic residual blocks, 16/32/64 channels) trained for 80 epochs with SGD (momentum 0.9, Nesterov, weight decay 2×10^{-4}), batch size 128, standard crop-and-flip augmentation, and a cosine learning-rate schedule (peak 0.1, five-epoch linear warmup). The only difference between the runs is the per-example loss weight. CE uses uniform weights throughout. CB applies the effective-number class-balanced weight $w_c \propto (1 - \beta)/(1 - \beta^{n_c})$ with $\beta = 0.9999$, renormalized to average to one across classes, from initialization. DRW trains with uniform weights until epoch 60 and switches to exactly the CB weights for the final twenty epochs (Cao et al., 2019); no other hyperparameter changes at the switch. The robustness matrix of Table 10 repeats this recipe verbatim at seeds 1 and 2 (ratio 100) and at ratios 10 and 50 (seed 0), re-drawing the long-tailed subset with the configuration’s own seed and initializing all three arms of a configuration identically (experiments/colab_cifar_multiseed.py). The post-hoc LA arm is not trained: it rescales the baseline’s test probabilities by the inverse training class prior, $q_{LA}(y | x) \propto q_{CE}(y | x)/\pi_{\text{train}}(y)$, renormalized.

J Sensitivity bounds on the reported estimates

What the bound says about this paper’s own estimates. Table 3 reports both forms of the bound on the Visual Genome pair of Section O.2, together with the coarsening from the class-pair to the subject-only cell, which instantiates Eq. (11) exactly with $Z = \text{object class}$. The free bound is directionally correct and, at $K = 50$, still 1,887 times the actual coarsening bias, so any $\bar{\rho}$ an analyst would plausibly declare certifies a limit far above what is observed. That looseness is a property of a worst-case inequality applied to label probabilities that concentrate away from the extremes at which it binds, not a defect of this dataset — but it also means the free bound cannot support a claim about whether any particular reported $\widehat{\Delta R}$ would survive confounding.

The sharp per-cell form of Eq. (13), which replaces the worst-case sums by the cell’s own $\sum_y \text{Var}(H_X(y) | c)$ and $\sum_y p_y(c)(1 - p_y(c))$, can support such a claim, and it is less reassuring than the free bound. Computed by experiments/sensitivity_analysis.py from per-instance arrays for the same pair, it gives $B = 0.23284$ at $\bar{\rho} = 1$, so the *robustness value* — the smallest aggregate correlation at which the bound reaches the estimate — is

$$\rho^\dagger = \frac{|\widehat{\Delta R}|}{B} = \frac{0.01427}{0.23284} = 0.06127. \quad (31)$$

Table 3: Both forms of the Proposition 3 bound on the MLP-SPATIAL vs. MLP-CLASS pair, and the coarsening already reported in Table 8 ($K = 50$, $M = 2$ for the quadratic score). The empirical bias is the law-of-total-expectation aggregate of Eq. (11)’s covariance term across all subject-only cells. `experiments/sensitivity_bound_check.py` reproduces the free-bound rows from the committed ablation results and `experiments/sensitivity_analysis.py` the sharp rows from per-instance arrays.

Quantity	Value
$\widehat{\Delta R}$, $\phi = \text{subject-only}$	0.02181
$\widehat{\Delta R}$, $\phi = \text{class-pair}$	0.01439
Empirical coarsening bias	0.00742
Free bound, $M\sqrt{K-1}$, at $\bar{\rho} = 1$	14.00
Minimum $\bar{\rho}$ for the free bound to reach the bias	0.00053
Sharp bound B at $\bar{\rho} = 1$	0.23284
Sharp bound at $\bar{\rho} = 0.1$	0.02328
Robustness value $\rho^\dagger$	0.06127

An unrecorded covariate whose aggregate correlation with the within-cell gain and the within-cell label distribution reaches 0.061 is enough to account for the whole of that $\widehat{\Delta R}$. At the $\bar{\rho} = 0.1$ that this section suggests as a plausible elicited value, the sharp bound is 0.02328, which *exceeds* the estimate. We therefore do not claim that a confounder would have to be implausibly strong to explain these Visual Genome gains away; on this evidence a modest one would suffice, and the reader should treat Section O.2’s covariance gains as conditional on the declared ϕ being close to complete. The flagship audit of Section 4.2 is not exposed in the same way, because its reported $\widehat{\Delta R}$ is indistinguishable from zero and a sensitivity bound on a null result changes nothing about it.

K Conditions for the limit theorem

Two things about condition (iv) deserve saying plainly rather than in a footnote. It is what licenses treating the per-cell plug-ins $\widehat{\Delta R}_c$ as consistent for ΔR_c in the variance argument, and a cell’s size is not controlled by the aggregate identified count, so this needs its own divergence requirement. But it also forces every cell to be eventually identified, hence $N_*/N \rightarrow 1$: an earlier version of this theorem additionally assumed an identified fraction converging to some $\kappa \in (0, 1]$, which condition (iv) makes vacuous, and under which the identified-subpopulation estimand θ would in any case coincide with the unconditional target. We have therefore dropped that assumption rather than state two conditions that cannot both bind. The price is that the theorem does not describe the regime the paper’s own applications sit in, where θ is a random, N -dependent target because coverage is short of one (99.0% on VG150, 98.9% in Section 4.2). What the theorem licenses is the interval’s form and its variance estimator; what licenses reading the reported intervals at finite N is the coverage simulation of Section 4.1, and Section 5 states the gap between the two rather than eliding it.

The proof (Section F) combines the exact linearization of Section G with a Lindeberg–Feller central limit theorem for the independent-across-cluster sum of mean-zero, uniformly bounded terms (Serfling, 1980; van der Vaart, 1998); boundedness (i) makes the Lindeberg condition automatic given (ii), and consistency of $\widehat{\sigma}^2$ follows from the weak law of large numbers applied to each plug-in $\widehat{\Delta R}_c$, $\widehat{\Delta R}$ together with continuity of the variance functional. One step needs more care than that section’s per-cell statement supplies: a per-cell Hájek remainder of order $o_p(n_c^{-1/2})$ does not aggregate to $o_p(N_*^{-1/2})$ at the dataset level when the number of cells grows, since summing C such remainders admits a $\sqrt{C}$ inflation. The repair does not need a new condition on cell sizes, only that the cell count be negligible relative to the identified sample, $\#\{c : n_c \geq 2\} = o(N_*)$: the second-order degenerate term of a two-sample U-statistic is $O_p(n_c^{-1})$ rather than $o_p(n_c^{-1/2})$, and $\sum_c n_c \cdot O_p(n_c^{-1}) = O_p(C) = o_p(N_*)$ gives the dataset-level rate directly. We state that condition explicitly; on VG150 it holds comfortably ($C = 6,346$ against $N_* = 227,337$). Condition (ii) is what

rules out a single cluster dominating the sum – an image contributing a positive fraction of all relations, for instance – and is a standard requirement for cluster-robust inference generally, not specific to this estimator. This theorem licenses reading the image-clustered intervals reported throughout Section 4 as asymptotically valid in the idealized regime where every eligible cell’s size grows without bound. That regime is not attained in-sample: VG150’s eligible cells are heavy-tailed, and 2,286 of the 6,346 eligible class-pair cells hold fewer than five examples (Table 8). The proportion is smaller than earlier drafts of this paper asserted — those cells carry 2.7% of identified relations, and the median eligible cell holds appreciably more than the minimum — but a third of the cells still sit where condition (iv)’s divergence is plainly hypothetical. The reported 94.0% empirical coverage against a nominal 95% (Section 4.1), not this theorem directly, is therefore the evidence that the interval remains a good approximation well short of the idealization.

L Choice of the grouping variable

Fineness is not a total order, and the guidance needs that qualification. The advice to prefer the finest defensible ϕ is sound against *coarsening*, which is what Proposition 2 governs. It gives no guidance at all between two partitions of the same granularity, neither of which coarsens the other — and there the estimand can differ without limit. Four cases suffice. Take two labels, q_0 uninformative at $(\frac{1}{2}, \frac{1}{2})$, and a new model that is confidently right on all four: $q_1 = (0.9, 0.1)$ on the two cases labelled 1 and $(0.1, 0.9)$ on the two labelled 2. The total gain is $\widehat{\Delta T} = +0.480$ under every partition, and every partition below is fully identified at 100% coverage:

declared ϕ	cells	$\widehat{\Delta P}$	$\widehat{\Delta R}$
$\{1, 3\}, \{2, 4\}$ (labels differ within each cell)	2×2	-1.120	+1.600
$\{1, 2, 3, 4\}$ (one cell)	1×4	-0.587	+1.067
$\{1, 2\}, \{3, 4\}$ (labels agree within each cell)	2×2	+0.480	0.000

The first and third partitions are equally fine, both fully identified, and they disagree about whether this model pair’s improvement is entirely case-level or entirely transported. The third is the degenerate case the paper already warns about — grouping by something that determines the label leaves each cell label-homogeneous and drives $\widehat{\Delta R}$ to zero algebraically — but the warning was previously attached to the choice $\phi = Y$, and the point here is more general and less comfortable: any ϕ correlated with the label inside a cell partially degenerates the estimand, “finest defensible” cannot detect it, and Proposition 2 is silent because neither partition is a coarsening of the other. What the analyst actually needs to declare, and what we recommend reporting alongside ϕ , is the within-cell label heterogeneity that remains: a partition whose cells are nearly label-homogeneous cannot identify much of anything, and the coverage statistic does not reveal it. This is a limitation of the construction rather than of any application of it, and we do not have a procedure that resolves it.

A recalibration-invariant alternative, and what it would cost. Because ΔR is a covariance between forecast values, its sensitivity to recalibration is structural rather than incidental, and a reader may reasonably ask why we do not use a rank-based within-cell contrast instead — a within-cell paired AUC difference, say, for which DeLong et al. (1988) already supply the correlated-U-statistic variance. Such a quantity would be invariant to any monotone rescaling of either model and would dissolve the confound this paper spends a protocol on. We think the trade is worth stating and is not obviously in our favour on every axis. What a rank-based contrast gives up is the exact identity: it is not a component of the score difference, so it cannot be added to anything to recover ΔT , and a benchmark reporting it learns how the models rank cases but not how the reported score gain decomposes. The construction here buys exact additivity with respect to a declared proper score, and pays for it with scale dependence. A rank-based diagnostic is the natural companion rather than a competitor, and reporting both — one exact and scale-dependent, one invariant and non-additive — is what we would now recommend to a benchmark that could choose.

Earlier versions of this corollary bounded $\sum_y \text{Var}(b_y \mid c)$ by Popoviciu’s inequality applied label by label, giving $K/4$ and hence $|\rho|KM/2$. That discards Eq. (14): the b_y are not K unconstrained quantities in $[0, 1]$

Table 4: The paper’s vocabulary, and what it corresponds to in standard terms.

Term used here	Standard counterpart	Meaning
grouping variable ϕ	stratifying / conditioning variable	a discrete function of the input, declared before the audit, whose levels the labels depend on
cell c	level of ϕ ; stratum	the set of test cases sharing one value of ϕ
label transport	within-stratum relabelling; permutation within blocks	replacing a case’s label with another case’s label from the same cell
total gain ΔT	paired score differential	the observed mean score of the new model minus that of the old
transported gain ΔP	no exact counterpart; a mean-forecast-fit term minus a dispersion term (Proposition 1), <i>not</i> a measure of prior fit	the part of ΔT that survives label transport
within-group covariance gain ΔR	the covariance term of a covariance rearrangement of a proper score (Yates, 1982); twice a classical resolution difference <i>only</i> under within-cell calibration	the part destroyed by transport; a within-cell prediction–label covariance
coverage	identified fraction	share of test cases in cells of size ≥ 2 , the only cells in which ΔR is identified
calibration-matched	temperature-scaled on held-out data	both models rescaled to comparable confidence before the audit

but a point on the simplex, and $K/4$ is unattainable for $K > 2$. At VG150’s $K = 50$ and $M = 2$ the corrected free bound is $14.00|\rho|$ rather than $50.00|\rho|$, a factor of 3.57.

An analyst willing to bound $|\rho| \leq \bar{\rho}$ – how strongly an unrecorded variable could jointly move a model’s within-cell score gain and the within-cell label distribution – obtains a numeric limit on how far $\widehat{\Delta R}_c$ can be from what a fully adjusted audit would find, without observing or modeling Z , in the spirit of Rosenbaum (2002) and recent partial-correlation sensitivity models for unmeasured confounding (Cinelli & Hazlett, 2020). Because $\bar{\rho}$ is a correlation rather than a domain-specific odds or risk ratio, it can be elicited the same way across benchmarks.

Both forms of the bound are evaluated on this paper’s own Visual Genome estimates in Section J. The short version is that the free bound is directionally correct but far too loose to certify anything, while the sharp per-cell form is *less* reassuring than the free one: an unrecorded covariate whose aggregate correlation with both channels reaches 0.061 would account for the entire covariance gain reported in Section O.2. We therefore do not claim those gains are robust to confounding, and Section 4.2’s audit is unaffected only because its estimate is already indistinguishable from zero.

Section K discusses what condition (iv) costs — it forces every cell to be eventually identified, so the theorem does not describe the finite-sample regime our own applications sit in — and states the additional condition on the number of cells that the dataset-level remainder requires. What licenses reading the reported intervals at finite N is the coverage simulation of Section 4.1, not this theorem.

M Terminology

N Composition of the transported gain

The component that survives transport is easy to misname. It is tempting to call ΔP a prior-fit gain, since transport replaces a case’s label by an independent draw from its cell; several earlier drafts of this paper did exactly that. The following identity shows the name is wrong, and says what the quantity is instead.

Only the first bracket is prior fit: it is positive exactly when the new model’s average within-cell forecast sits closer to the cell’s label frequencies. The second has nothing to do with the prior. It rewards the new model for making *less* dispersed predictions inside the cell, at a fixed mean forecast, so ΔP can be large and positive when neither model’s prior fit has changed at all. One cell suffices: with two labels and frequencies $(\frac{1}{2}, \frac{1}{2})$, let q_0 alternate between $(1, 0)$ and $(0, 1)$ in a pattern uncorrelated with the label and let q_1 predict $(\frac{1}{2}, \frac{1}{2})$ throughout. Both mean forecasts equal the cell frequencies exactly, so the first bracket is zero, and the whole observed gain of $\frac{1}{2}$ is transported, produced entirely by the collapse in dispersion. The channel is therefore the reliability-and-sharpness channel of the classical partition, not a prior-fitting channel, which is why we name it for the operation. Two consequences: a claim that some gain is “mostly prior-fitting” does not follow from a large ΔP alone, and Eq. (8) explains why confidence matching moves ΔP so sharply, since a temperature change is almost purely a change in dispersion.

O Further empirical studies

This section reports the two studies summarized in Section 4.5 and the granularity, score and cell-size sensitivity analyses referred to there. Each applies the estimator of Section 3 without modification; only the declared grouping variable ϕ changes.

O.1 Controlled predictors on VG150 in full

We work with Visual Genome Krishna et al. (2017) in the PredCls regime Xu et al. (2017), where ground-truth boxes and object labels are supplied and the model has only to predict one of 50 predicate labels for each ordered object pair. The predictors of this section are audited on a VG150-*style* dataset we build directly from the released relationship annotations: the 150 most frequent object classes and 50 most frequent predicates form the vocabulary, and images are split 70/30 into a pool for fitting the compared predictors and a pool on which the audit runs. We use this reconstruction rather than the canonical VG150 release because these predictors are trained here from annotations alone; Section 4.2 audits *released third-party checkpoints* on the canonical split itself, so both settings appear in the paper and neither is asked to stand for the other. The reconstruction provides 532,987 training and 229,605 test relations, drawn from 65,228 and 27,955 images respectively, and the cell throughout is $\phi = (\text{subject class}, \text{object class})$, two real examples of which appear in Figure 3. Of the 8,614 cells present in the test split, 6,346 contain at least two relations and are therefore identified; the remaining 2,268 are singletons, which WAGER excludes and reports rather than smoothing over, leaving 227,337 relations (99.0%) under audit.

The five predictors are deliberately constructed so that each exposes a different kind of information. FREQ is the training-count distribution conditioned on the class pair, with add-one smoothing and a subject-marginal and then global backoff for class pairs unseen in training; FREQ+OVERLAP adds to it a single binary box-overlap indicator. MLP-CLASS consumes fixed subject and object class embeddings, which makes it a function of ϕ alone; MLP-SPATIAL extends those embeddings with 12 normalized box-geometry features covering relative centre, scale, aspect, intersection, containment and IoU, while MLP-SPATIAL+ both widens the network and trains longer (Section I lists every architecture and schedule), so its increment over MLP-SPATIAL reflects capacity and optimization jointly rather than capacity alone. Each is trained once on the training split, and WAGER thereafter consumes nothing but their cached test distributions, scored quadratically with image-clustered intervals and 499 within-cell cyclic randomizations, whose smallest attainable p -value is $1/500 = .002$; table entries of .002 are at this granularity floor and should be read as $p \leq .002$. With confidence intervals as the primary uncertainty statement and every significant effect many standard errors from zero, we do not additionally adjust the diagnostic p -values for the number of reported comparisons.

O.2 Gain attribution for the controlled predictors

The central result is collected in Table 6 and Figure 4. MLP-CLASS improves on FREQ by 0.03386 of quadratic score, yet because its gain vector is constant inside every class-pair cell, the estimator assigns that entire improvement to prior transport and precisely nothing to instance covariance. The zero is worth

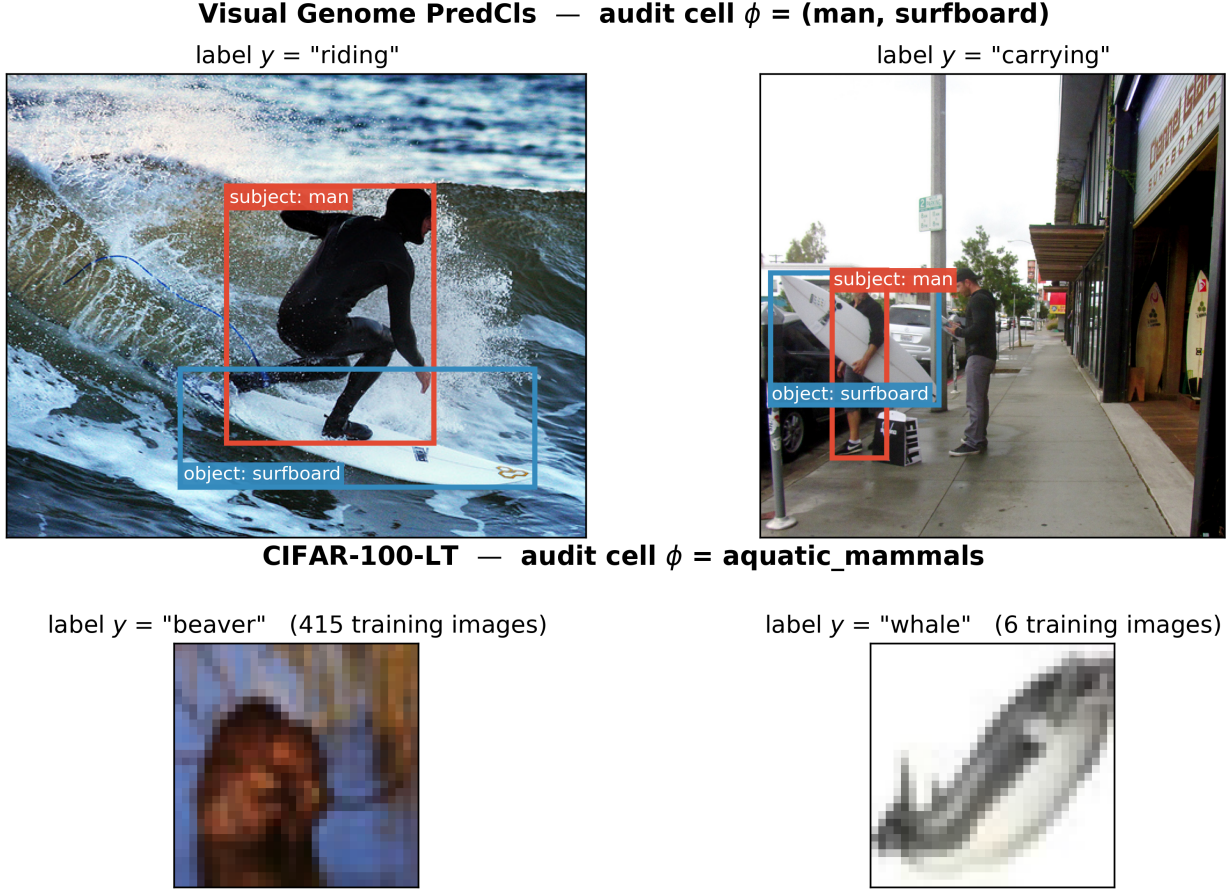


Figure 3: **What a cell contains, on both benchmarks.** Each row shows two real test examples that share the same declared grouping variable ϕ but carry different labels – precisely the configuration within-cell transport exploits. *Top*: two Visual Genome relations from the cell $\phi = (\text{man}, \text{surfboard})$. The class pair is identical and therefore cannot distinguish “riding” from “carrying”; only the image content can, which is what ΔR is designed to credit. Boxes are the annotated subject (red) and object (blue) regions that MLP-SPATIAL turns into geometry features and MLP-VISUAL crops and encodes with CLIP. *Bottom*: two CIFAR-100 test images from the superclass $\phi = \text{aquatic_mammals}$, drawn from opposite ends of the long-tailed training distribution (415 versus 6 training images). Crossing the labels within a row preserves the cell and its label histogram while destroying which prediction belongs with which instance.

Table 5: Frozen-model top-1 accuracy on VG150 PredCls. Similar accuracies do not reveal whether the difference is transported.

Model	Accuracy
FREQ	0.6564
FREQ+OVERLAP	0.6545
MLP-CLASS	0.6554
MLP-SPATIAL	0.6639
MLP-SPATIAL+	0.6674

dwelling on: it follows algebraically from the construction rather than from a learned baseline happening to pass a sanity check.

Table 6: WAGER decomposition on 227,337 identified VG150 PredCls relations. Gains use the quadratic score. CI is image-cluster robust; p is the 499-shift relation-level randomization diagnostic (granularity floor .002). “–” denotes an undefined share because total gain is negative. The MLP-CLASS row’s zero-width interval is exact by construction, not a sampling statement: a gain vector constant within every cell makes the antisymmetric kernel identically zero (Theorem 1), so no variability exists for the interval to measure.

New model	Old model	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$ (95% CI)	$\hat{\rho}$	p
FREQ+OVERLAP	FREQ	−0.01229	−0.01430	0.00202 [0.00168, 0.00235]	–	.002
MLP-CLASS	FREQ	0.03386	0.03386	0.00000 [0.00000, 0.00000]	0.00	.724
MLP-SPATIAL	FREQ	0.04270	0.02832	0.01439 [0.01373, 0.01504]	0.34	.002
MLP-SPATIAL+	FREQ	0.04644	0.02885	0.01760 [0.01686, 0.01834]	0.38	.002
MLP-SPATIAL	MLP-CLASS	0.00885	−0.00554	0.01439 [0.01373, 0.01504]	1.63	.002
MLP-SPATIAL+	MLP-SPATIAL	0.00374	0.00053	0.00321 [0.00289, 0.00353]	0.86	.002

Adding box geometry changes the picture. MLP-SPATIAL gains 0.04270 over FREQ, of which 0.02832 survives label transport and 0.01439 is within-group covariance (CI [0.01373, 0.01504]). More revealing still is the direct comparison against MLP-CLASS, where the total gain falls to 0.00885 only because the transported channel deteriorates by 0.00554 even as the covariance gain from spatial features rises by 0.01439. The resulting covariance share of 1.63 records a genuinely useful instance-specific change that the aggregate score partly conceals behind a worse prior component, and it is exactly the configuration that a ratio confined to $[0, 1]$, or thresholded at 0.5, would misdescribe.

The remaining two comparisons bracket the interesting range. MLP-SPATIAL+ adds 0.00374 over MLP-SPATIAL, nearly all of it covariance gain at 0.00321 (CI [0.00289, 0.00353]), whereas FREQ+OVERLAP supplies the opposite caution: its overlap indicator does create a small positive covariance gain, but the accompanying negative prior term is larger, so the model ends up worse overall. Using image-derived information is thus never treated as an improvement in itself; both channels are reported, signs included.

Reading the magnitudes. Quadratic-score units are not ones most readers calibrate by eye, so it is worth anchoring them in the rank-based metrics this literature reports. Where both channels point the same way the correspondence is straightforward: MLP-SPATIAL-S’s $\widehat{\Delta R} = +0.01023$ over MLP-CLASS-S accompanies +0.51 points of top-1 predicate accuracy, +0.60 of mean reciprocal rank and +0.86 of recall@5, so in these data a hundredth of quadratic-score gain is worth roughly half an accuracy point. The covariance channel is not, however, a re-description of any rank metric, and the real-pixel comparison of Section 4.4 is where that becomes visible: there $\widehat{\Delta R} = +0.00641$ while top-1 accuracy falls by 3.07 points and recall@5 by 2.21. Prior-matching both models before scoring narrows the accuracy gap to 2.17 points without closing it. The rank metrics aggregate the two channels into one number; when the channels carry opposite signs, as they do here, no single such number can represent both, which is precisely the situation the decomposition exists to report.

Reading $\widehat{\Delta T}$ as a comparative forecast test. Corollary 3 identifies $\widehat{\Delta T}$ and its image-clustered interval in Table 6 as a grouped Diebold–Mariano statistic and interval, on the identified subsample, for each pair’s paired score differential, so every row already reports what that classical test would report. What it does not report is the next two columns: whether MOTIFS-TDE (Section 4.2) or MLP-SPATIAL beat their baseline for a reason a frequency-only remodeling of the same benchmark could not reproduce, which is the question $\widehat{\Delta P}$ and $\widehat{\Delta R}$ answer and DM/GW do not ask.

0.3 Real pixels against a geometric proxy, in full

None of the predictors above ever sees an image: they are functions of the class pair and, for MLP-SPATIAL(+), of box coordinates recorded in the annotation. Since box geometry is only a proxy for image content, a sceptical reader may ask whether the covariance channel responds to pixel evidence or merely

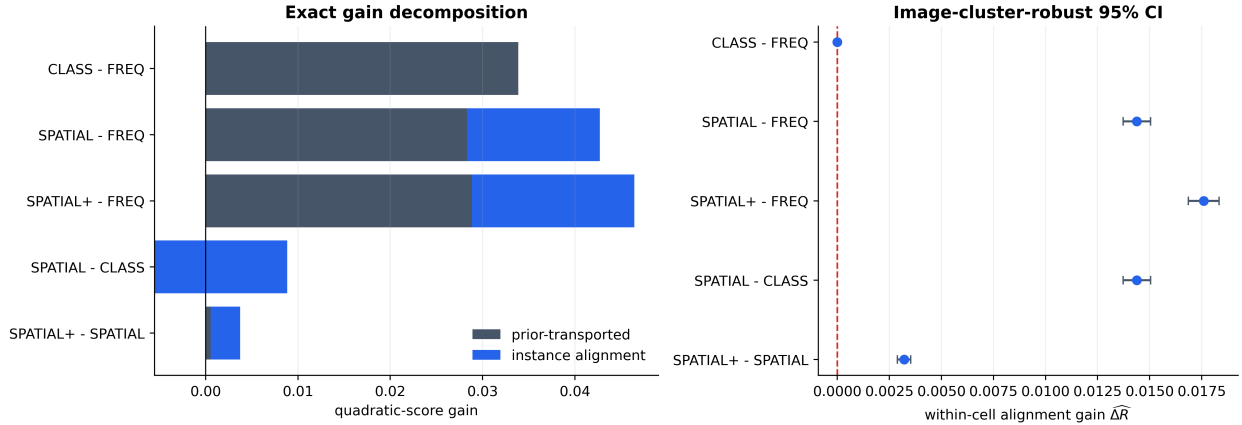


Figure 4: **Direct model-pair attribution.** Left: exact decomposition of each positive gain into transported and within-group covariance components (the negative prior term for SPATIAL–CLASS extends left of zero). Right: primary covariance estimates with image-cluster-robust intervals.

to that proxy. Figure 3 sharpens the question — the cell $\phi = (\text{man}, \text{surfboard})$ holds both “riding” and “carrying”, and what distinguishes them is what the photograph shows.

MLP-VISUAL answers it. Subject and object regions are cropped from the original image and each encoded by a frozen CLIP ViT-B/32 encoder (Radford et al., 2021); the two 512-dimensional embeddings are concatenated with the class embeddings MLP-CLASS already uses and passed to a small trained head. Because CLIP is never fine-tuned, whatever covariance the model gains reflects information its pretraining already recovered from pixels rather than representation learning on Visual Genome.

Encoding every training relation with CLIP is expensive, so MLP-VISUAL must be trained on a subsample; setting a subsampled model against the full-data MLP-SPATIAL would confound feature content with training-set size. All three compared predictors are therefore retrained on one fixed, seeded subsample of 100,000 of the 532,987 training relations, written MLP-CLASS-S, MLP-SPATIAL-S and MLP-VISUAL-S, so any gap between them isolates the features. The audit still uses the full 229,605-relation test split with the same class-pair ϕ and image-clustered intervals.

Result. By any aggregate measure the CLIP model is the worst of the three: top-1 accuracy 0.6161 against 0.6467 for MLP-SPATIAL-S and 0.6416 for MLP-CLASS-S, and a quadratic score 0.04655 below MLP-SPATIAL-S. An evaluator reading either number would discard it.

The decomposition contradicts that reading on the axis that matters. Against MLP-SPATIAL-S the CLIP model loses 0.05296 of transported gain while *gaining* 0.00641 of covariance (95% CI [0.00514, 0.00769], randomization $p = .002$): its entire deficit sits in the transported channel and its within-cell discrimination is significantly better, not worse. Against the class-only baseline the point is plainer still — pixels buy 0.01665 of covariance where box geometry buys 0.01023, with well-separated intervals. Real image content carries more case-specific signal than the geometric proxy standing in for it.

A plausible mechanism, stated as hypothesis rather than finding: the 1024 CLIP dimensions dominate the 64 dimensions of class embedding the head also receives, so the model fits VG150’s class-pair frequency structure markedly less well than a predictor built around it. A benchmark whose headline metric rewards transported gain will punish this model, and the aggregate numbers duly do. What the decomposition adds is that the punishment is not for failing to use the image. Two falsification checks, reported in Section I, support that reading: correcting a model’s within-cell prior towards the training histogram registers as almost pure transported movement (+0.02401 against +0.00111 of covariance), as a correction carrying no case-level information must, and applied to the CLIP model it halves the aggregate deficit (−0.04655 to −0.02143) while the covariance advantage persists (+0.00752, and +0.00663 when both models are corrected) and survives confidence matching (+0.00467, CI [+0.00314, +0.00621]).

Table 7: Matched-subsample real-pixel study on the full VG150 test split (coverage 99.0%, as in Table 6). All three predictors are trained on the same 100,000 relations and differ only in features. CI is image-cluster robust on $\widehat{\Delta R}$; p is the 499-shift randomization diagnostic.

New model	Old model	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$ (95% CI)	p
MLP-SPATIAL-S	MLP-CLASS-S	0.00503	−0.00520	0.01023 [0.00967, 0.01079]	.002
MLP-VISUAL-S	MLP-CLASS-S	−0.04152	−0.05816	0.01665 [0.01534, 0.01796]	.002
MLP-VISUAL-S	MLP-SPATIAL-S	−0.04655	−0.05296	0.00641 [0.00514, 0.00769]	.002

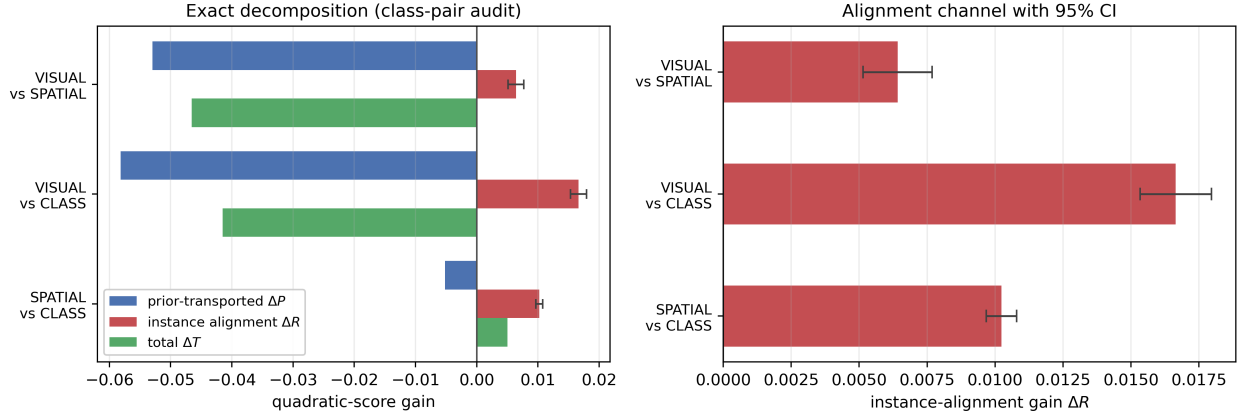


Figure 5: **Pixels beat geometry on covariance while losing on the aggregate.** Left: the exact decomposition for the three matched-subsample pairs. Both comparisons involving the CLIP model have strongly negative totals (green) driven entirely by the transported channel (blue), while their covariance channel (red) is positive in every case. Right: the covariance channel alone, with 95% image-clustered intervals. Replacing box geometry with real image content raises within-group covariance from 0.01023 to 0.01665 over the same class-only baseline.

0.4 Sensitivity analyses

For MLP-SPATIAL versus MLP-CLASS, the conclusion survives every planned sensitivity check (Table 8). Changing from the quadratic to a probability-floored log score changes scale (0.01439 to 0.06536) but not sign or significance. The coverage column matters for reading the rows against each other, and earlier versions of this table omitted it. Each row targets the identified subpopulation its own setting defines, so the minimum-cell-size rows are not four estimates of one quantity: raising the minimum from 2 to 20 drops coverage from 99.0% to 86.7% and changes the estimand along with the estimator. That the estimate moves so little across them (0.01439 to 0.01225) is therefore mild evidence of stability rather than a sensitivity analysis holding the target fixed.

Coarsening ϕ from the class pair to subject class increases the credited covariance to 0.02181. Proposition 2 identifies why: the coarser cell’s estimate equals the class-pair estimate’s conditional average plus a between-class-pair covariance term, and that term is positive here, meaning some of the credited gain reflects class-pair-level structure that the finer audit had already correctly assigned to ΔP . This is precisely the sense in which the class-pair audit is the conservative choice: it is not that coarsening is wrong, but that it cannot be assumed neutral, and only the finer partition’s estimate isolates covariance gain net of every prior regularity expressible in ϕ . The same class-pair-versus-subject-only pair also instantiates Proposition 3 directly, with the subject class playing the role of the unrecorded confounder Z : Table 3 in Section J reports that the resulting bias of 0.00742 is comfortably inside the Cauchy–Schwarz bound, though the bound is loose enough at $K = 50$ that it would remain satisfied for a $\bar{\rho}$ three orders of magnitude smaller than one. Raising the

Table 8: Sensitivity of the MLP-SPATIAL vs. MLP-CLASS decomposition to score, audit granularity, and minimum cell size. CI is image-cluster robust on $\widehat{\Delta R}$.

Factor	Setting	$\widehat{\Delta T}$	$\widehat{\Delta R}$ (95% CI)	Cells	Cov.
Score	Brier (def.)	0.00885	0.01439 [0.01373, 0.01504]	6346	99.0%
Score	Log (floored)	0.04297	0.06536 [0.06294, 0.06779]	6346	99.0%
Cell ϕ	class pair (def.)	0.00885	0.01439 [0.01373, 0.01504]	6346	99.0%
Cell ϕ	subject only	0.00943	0.02181 [0.02079, 0.02283]	150	100.0%
Min. cell size	2 (def.)	0.00885	0.01439 [0.01373, 0.01504]	6346	99.0%
Min. cell size	5	0.00772	0.01395 [0.01329, 0.01462]	4060	96.3%
Min. cell size	10	0.00683	0.01330 [0.01262, 0.01398]	2744	92.6%
Min. cell size	20	0.00570	0.01225 [0.01155, 0.01295]	1757	86.7%

minimum cell size from 2 to 5, 10, and 20 yields 0.01395, 0.01330, and 0.01225; all cluster-robust lower limits remain positive. The procedure itself has no smoothing strength, split fraction, or decision threshold to tune.

O.5 Long-tailed image classification

Sections 4.3–4.4 are all instances of one task (predicate classification on an ordered object pair). Section 2.4 argued that long-tailed image classification poses a structurally identical question: a class-frequency prior can be fit better without any per-instance improvement. We test this directly on CIFAR-100-LT (Cui et al., 2019), an exponential-imbalance (ratio 100) subsample of CIFAR-100’s training split with the original, balanced test split retained for evaluation. We train ResNet-32 (He et al., 2016) classifiers that differ only in their per-class loss weighting: a vanilla cross-entropy baseline (CE); class-balanced effective-number re-weighting applied from initialization (CB) (Cui et al., 2019); and the same re-weighting deferred to the final twenty epochs (DRW) (Cao et al., 2019). Optimizer, schedule, augmentation, epoch budget, and seed are identical across runs (Section I).

Choosing ϕ . A declared grouping variable has to predict the label without determining it, and this second benchmark makes the constraint unusually easy to violate. Were ϕ taken to be the fine class label itself, every cell would be label-homogeneous, within-cell transport would have nothing to cross, and $\widehat{\Delta R}$ would come out identically zero – an algebraic artifact wearing the appearance of a finding. CIFAR-100’s own 20 coarse superclasses avoid this, each gathering five fine labels into a cell of 500 balanced test images, which is structurally the situation of a VG class-pair cell holding several predicates (Figure 3). Alongside the superclass audit we report two further declared priors: the trivial coarsening to a single global cell, which audits against the global class marginal and so instantiates Proposition 2 in a second domain, and a partition into head, medium and tail training-frequency tiers. CIFAR-100 test images are independent, unlike the several relations VG draws from one photograph, so no cluster grouping is required and the plain Hájek interval of Section 3.9 applies directly. One semantic point matters in this domain: transport uses each cell’s *test* label histogram, which on CIFAR-100-LT is balanced. A positive ΔP here therefore records movement *toward* the balanced test marginal – undoing the training-frequency skew – which is precisely the adjustment long-tailed evaluation is designed to reward; a transported gain is a description of where score was earned, not an accusation (Section 5).

Result. The three classifiers reach top-1 accuracies of 0.3662 (CE), 0.2627 (CB) and 0.3874 (DRW), so re-weighting from initialization degrades the classifier substantially while the deferred schedule delivers the improvement it was designed to produce. Table 11 decomposes both gains against the superclass audit.

Of the two, the class-balanced run is the more instructive, precisely because its aggregate score conceals almost everything of interest. On the quadratic score it gains +0.00143 over the baseline, a difference small enough that a reader working from the headline number alone would reasonably conclude that re-weighting had changed very little. The decomposition tells another story: within superclasses the transported channel improves by +0.19713 while within-group covariance deteriorates by −0.19569 (95% CI [−0.20728, −0.18411]), so the near-zero total is not a small effect but two large ones that happen to cancel.

Table 9: Raw and calibration-matched decompositions on CIFAR-100-LT at the superclass audit. Temperatures are fitted by likelihood on a held-out calibration half of the test split; every decomposition uses the disjoint audit half; entries are means (sd) over twenty random splits. The final row is the control: recalibrating the baseline alone carries no instance information yet moves both channels substantially.

Comparison	Regime	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$
CB vs CE	raw	+0.00208 (0.00642)	+0.19635 (0.00369)	−0.19427 (0.00632)
CB vs CE	calibration-matched	−0.12912 (0.00373)	+0.11079 (0.00372)	−0.23991 (0.00491)
DRW vs CE	raw	+0.06712 (0.00865)	+0.04258 (0.00463)	+0.02454 (0.00685)
DRW vs CE	calibration-matched	+0.02173 (0.00275)	+0.00216 (0.00410)	+0.01957 (0.00329)
CE recalibrated vs CE	control	+0.20495 (0.00420)	+0.37960 (0.00268)	−0.17464 (0.00455)

A calibration-matched regime. A channel split of that shape could, however, be produced by confidence scaling alone. Under the quadratic score the covariance channel is a covariance between probability movements and labels (Eq. (7)), and a monotone recalibration moves it: softening an overconfident model transfers score mass from the covariance channel to the transported channel while adding no information about any individual image. The baseline invites exactly this concern, assigning a mean maximum probability of 0.764 while being correct on only 0.3662 of the test set. We therefore repeat the audit in a calibration-matched regime: each model’s temperature is chosen by held-out likelihood on a calibration half of the test split ($T^* = 2.85$ for CE, 3.51 for CB, 2.61 for DRW), the decomposition is computed on the disjoint audit half, and the whole procedure is repeated over twenty random splits (Table 9). The regime’s necessity is quantified by its control row: recalibrating the baseline alone – a transformation carrying no instance information – moves the channels by +0.380 (prior) and −0.175 (covariance), so the raw split does mix calibration with discrimination.

The calibration-matched rows then separate the two schedules cleanly, and more sharply than the raw ones. CB’s covariance deficit does not shrink but widens, to −0.23991 (0.00491): the model re-weighted from initialization has genuinely lost within-superclass discrimination – a real degradation that the ten-point accuracy drop records independently – and its near-zero raw total reflects that loss offset by its softer, better-calibrated confidence. DRW inverts the reading its raw split suggests. Raw, its +0.06712 gain divides roughly two-thirds transported to one-third covariance gain; calibration-matched, the improvement is +0.02173 (0.00275) of which +0.01957 (0.00329) is within-group covariance while the transported channel is close to zero (+0.00216 (0.00410)). What survives calibration matching is precisely the component the deferred schedule was designed to buy – per-instance discrimination, concentrated on rare classes (Table 12) – while the apparently transported share of its raw gain was largely the calibration difference between the two runs. Neither an accuracy difference nor an aggregate score exposes either composition, which is the practical argument for reporting the two channels, in both regimes, instead of their sum.

Seeds, imbalance ratios, and a post-hoc arm. A decomposition computed from one trained pair carries only test-set sampling uncertainty, which is not the uncertainty relevant to a claim about a training *method*. We therefore retrained the triple at two further seeds at the primary ratio and at imbalance ratios 10 and 50, and added a fourth arm: post-hoc logit adjustment (LA), which divides the baseline’s predictions by the training class prior without retraining (Menon et al., 2021). Every configuration is audited in both regimes; Table 10 reports the calibration-matched covariance channel, which is the one the preceding paragraphs interpret.

Three things follow. First, the class-balanced run’s covariance loss is not a misconfiguration but a systematic function of imbalance severity: −0.069 at ratio 10, −0.244 at 50, and −0.251 (0.011) across three seeds at 100, tracking an accuracy penalty that grows from 2.3 to 10.4 points over the same range. Second, the deferred schedule’s covariance gain is positive at every ratio and in every seed, but its magnitude varies roughly fourfold across seeds at ratio 100 (+0.007 to +0.031). We therefore report the spread rather than a seed-level interval: the direction replicates, the magnitude is seed-sensitive, and the gap between the

Table 10: Calibration-matched covariance gain across seeds and imbalance ratios, with balanced test accuracy for each arm. LA is post-hoc logit adjustment applied to the baseline. The class-balanced deficit scales with imbalance severity; the deferred schedule’s gain is positive throughout but seed-sensitive in magnitude; the post-hoc correction lands in the covariance channel rather than the transported channel.

Ratio	Seed	Top-1 accuracy				$\widehat{\Delta R}$ vs. CE (calibration-matched)		
		CE	CB	DRW	LA	CB	DRW	LA
10	0	0.5388	0.5154	0.5447	0.5497	−0.06903	+0.00617	+0.02222
50	0	0.4256	0.3242	0.4399	0.4550	−0.24409	+0.01903	+0.03768
100	0	0.3662	0.2627	0.3874	0.3959	−0.23936	+0.01987	+0.03449
100	1	0.3614	0.2287	0.3762	0.3877	−0.26176	+0.00748	+0.02672
100	2	0.3687	0.2481	0.3968	0.4024	−0.25213	+0.03124	+0.03601
100, mean (sd)						−0.25108 (0.01124)	+0.01953 (0.01188)	+0.03241 (0.00498)

tight within-run intervals of Table 11 and this across-seed spread is exactly why a method-level claim needs replication rather than a single audit.

Third, LA is an instructive falsification of a natural expectation. A purely post-hoc prior correction might be expected to register as almost pure transported channel, as the within-cell prior matching of Section 4.4 does. It does not: calibration-matched, LA reads as almost entirely covariance gain (+0.022 to +0.038) with a transported channel near zero, while achieving the best balanced accuracy of any arm at every ratio. The reason is visible in the construction. Dividing by the *global* class prior is not a within-superclass histogram move, because fine classes inside a superclass differ in training frequency, and the correction it applies is multiplicative in each example’s own predicted probabilities — it changes rankings where the model already carried latent evidence. The two post-hoc corrections in this paper therefore land in opposite channels, and correctly so: transport separates adjustments that act on the cell’s label histogram from adjustments that act on individual predictions.

Where the covariance gain sits is visible in Table 12 and Figure 6 (raw regime, primary seed). For DRW it falls where the long-tail literature intends it to, reaching +0.09009 on few-shot classes and +0.06760 on medium-shot ones, while many-shot classes lose −0.07624 as the schedule shifts capacity away from the head. The class-balanced run reproduces that same ordering with every level displaced downward, its many-shot covariance collapsing to −0.41977. What the decomposition supplies, then, is not only how much of a long-tail gain is real but where in the frequency spectrum it was earned.

These numbers also provide an implementation check on real inputs. Computing the within-group covariance gain directly from the identity of Eq. (7), through a code path that shares nothing with the transport estimator, gives −0.19530; multiplying by the factor $n_c/(n_c-1)$ that Proposition 4 predicts, at the $n_c = 500$ of a balanced superclass cell, returns −0.195693 against the estimator’s −0.19569. The identities of Theorem 1 and Proposition 4 are algebraic, so this verifies the implementation rather than the theory; the point of the check is that two independent code paths agree to the fifth decimal on real trained models, not only on the synthetic populations used in the unit tests.

Coarsening behaves as Proposition 2 predicts. Moving from the 20 superclass cells to a single global cell changes the credited covariance from −0.19569 to −0.25430; the difference is the between-cell covariance term, which is negative here, so the coarse audit overstates the covariance loss by attributing to it structure that the superclass partition correctly assigns to ΔP . The finer partition is again the conservative reading, for the same reason as in Section O.4 and now in a second domain.

O.6 Long-tailed text classification

Sections O.2–O.5 test two vision tasks that share cached probability vectors and grouping variables grounded in class frequency or object geometry. To test the breadth Section 5 claims – any benchmark supplying two models’ full probability vectors, labels, and a declared discrete grouping variable – we apply the identical estimator, unmodified, to document category prediction on 20 Newsgroups (Lang, 1995). Neither the pre-

Table 11: WAGER decomposition on the 10,000-image balanced CIFAR-100-LT test split. Gains use the quadratic score; CI is the Hájek interval on $\widehat{\Delta R}$; p is the 499-shift randomization diagnostic, one-sided for a positive covariance gain, so the value of 1.000 on the CB rows records a covariance gain at the extreme negative end of its randomization distribution.

Comparison	Audit cell ϕ	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$ (95% CI)	p
CB vs CE	superclass (20)	0.00143	0.19713	-0.19569 [-0.20728, -0.18411]	1.000
CB vs CE	global (1)	0.00143	0.25573	-0.25430 [-0.26801, -0.24058]	1.000
CB vs CE	tier (3)	0.00143	0.25247	-0.25103 [-0.26453, -0.23754]	1.000
DRW vs CE	superclass (20)	0.06606	0.04206	0.02400 [0.01340, 0.03460]	.002
DRW vs CE	global (1)	0.06606	0.03522	0.03085 [0.01860, 0.04309]	.002
DRW vs CE	tier (3)	0.06606	0.03702	0.02904 [0.01702, 0.04107]	.002

Table 12: Covariance gain by training-frequency tier at the primary superclass audit. The deferred schedule concentrates its genuine per-instance improvement on rare classes; both methods lose covariance on the head.

Tier	n	CB vs CE	DRW vs CE	
		$\widehat{\Delta R}$	$\widehat{\Delta T}$	$\widehat{\Delta R}$
Few-shot (< 20)	3000	-0.03337	0.13785	0.09009
Medium-shot (20–100)	3500	-0.11075	0.10799	0.06760
Many-shot (> 100)	3500	-0.41977	-0.03740	-0.07624

dictors nor the grouping variable are visual here: documents replace images, and the declared ϕ is a topical grouping rather than a spatial or class-frequency one, so a favorable result cannot be attributed to anything specific to vision data.

We build a long-tailed training subset with the same exponential profile as CIFAR-100-LT (imbalance ratio 100, Section O.5), applied to 20 Newsgroups’ 20 fine categories, keeping the standard balanced test split (7,532 documents) untouched (Section I). The 20 categories fall into six conventional coarse topics – computing, recreation, science, politics, religion, and classifieds – which we use as the primary declared ϕ , the direct analogue of CIFAR’s coarse superclasses; a global (trivial) and a training-frequency-tier partition are reported alongside, as in Section O.5. Documents are represented by a TF-IDF vector reduced to 300 dimensions by truncated SVD, and the three compared classifiers – CE, CB, and DRW – follow the same three-arm protocol as Section O.5: an identical small MLP architecture and optimizer across runs, differing only in per-example loss weight (Section I). Test documents are independent units, so the plain Hájek interval of Section 3.9 applies directly, as for CIFAR.

Result. Table 13 decomposes both re-weighting schedules’ gain over the cross-entropy baseline (top-1 accuracies 0.3532 CE, 0.3960 CB, 0.3776 DRW). The two schedules land in opposite channels here, which is itself the finding worth reporting: on this task it is class-balanced re-weighting from initialization, not the deferred schedule, that buys genuine within-topic covariance gain (+0.04317, CI [0.03664, 0.04970], 37.9% of its total gain), while DRW’s comparably sized total gain (0.11490 vs. CB’s 0.11388) is 98.6% transported, its residual covariance statistically indistinguishable from zero at the superclass audit (+0.00155, CI [-0.00296, 0.00605], $p = .236$) and significantly negative once the coarser tier partition is used instead (-0.00901, CI [-0.01428, -0.00374]). This is the reverse of Section O.5’s image-classification finding, where CB’s near-zero total masked a large covariance-gain *loss* and DRW’s gain was genuinely covariance-driven. Nothing about the estimator or the construction of ϕ changed between the two applications; what changed is which training recipe, on which data, buys real per-instance discrimination – exactly the composition question an aggregate score or an accuracy difference alone cannot answer, now demonstrated on a domain with neither a class-frequency table nor a spatial prior.

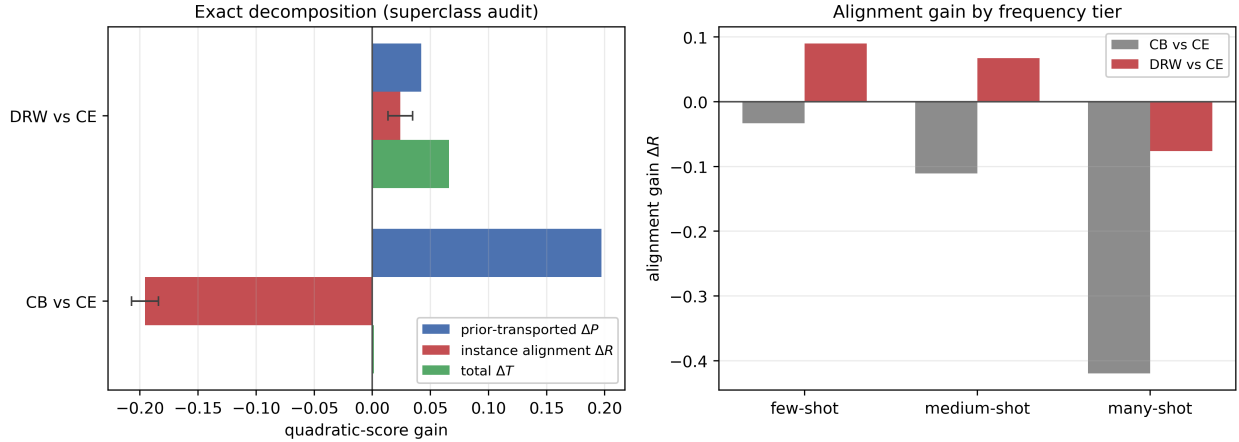


Figure 6: **The same attribution transfers to a second task.** Left: exact decomposition at the superclass audit. The CB total gain (green) is visually almost absent at $+0.00143$ while its two channels reach ± 0.2 – a near-zero aggregate score concealing two large opposing effects. DRW’s smaller total is genuinely split between both channels. Error bars are 95% Hájek intervals on $\widehat{\Delta R}$. Right: the covariance channel by training-frequency tier, showing that DRW’s genuine per-instance gain is concentrated on rare classes and negative on the head.

Table 13: WAGER decomposition on the 7,532-document balanced 20-Newsgroups-LT test split. Gains use the quadratic score; CI is the (unclustered) Hájek interval on $\widehat{\Delta R}$, documents being independent units; p is the 499-shift randomization diagnostic, one-sided for a positive covariance gain.

Comparison	Audit cell ϕ	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$ (95% CI)	p
CB vs CE	superclass (6)	0.11388	0.07071	0.04317 [0.03664, 0.04970]	.002
CB vs CE	global (1)	0.11388	0.07203	0.04185 [0.03381, 0.04990]	.002
CB vs CE	tier (3)	0.11388	0.07810	0.03578 [0.02825, 0.04331]	.002
DRW vs CE	superclass (6)	0.11490	0.11335	0.00155 [−0.00296, 0.00605]	.236
DRW vs CE	global (1)	0.11490	0.11809	−0.00319 [−0.00884, 0.00245]	.980
DRW vs CE	tier (3)	0.11490	0.12391	−0.00901 [−0.01428, −0.00374]	1.000

Coarsening moves both comparisons in the direction Proposition 2 allows without requiring: CB’s credited covariance falls from 0.04317 at the superclass partition to 0.04185 at the global cell and 0.03578 at the tier partition, while DRW’s changes sign twice, from a small positive residual (superclass) to a small negative one (global) to a significant negative one (tier). None of these differences overturns CB’s conclusion, but DRW’s illustrates exactly why the finest defensible partition is the conservative default (Section 3.5): a reader stopping at the coarser tier partition would report a significant covariance *loss* for a method whose finer-grained residual is not distinguishable from zero.

Table 14 breaks the superclass-audit covariance down by training-frequency tier for both schedules; the two methods agree in shape even though their superclass totals diverge sharply in composition, each gaining covariance on rare and medium-frequency categories and losing it on frequent ones, the same qualitative signature Section O.5 reports for image classification.

This third domain uses a single trained pair per schedule rather than the multi-seed, calibration-matched protocol Section O.5 applies to CIFAR-100-LT; extending that fuller protocol to text is the natural next replication, not a step this section claims to have taken.

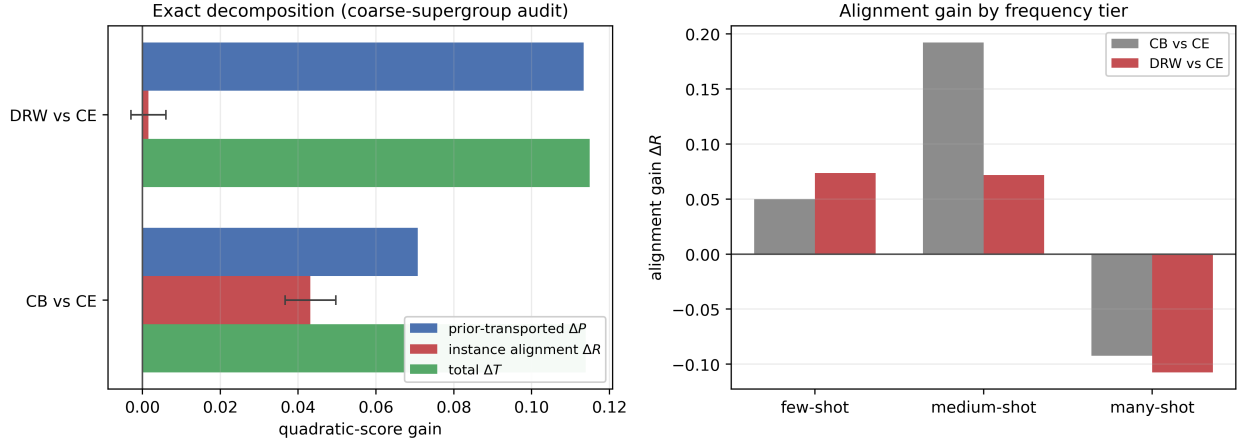


Figure 7: **A third domain, with neither a spatial nor a class-frequency prior.** Left: exact decomposition at the coarse-topic audit; CB’s total gain is genuinely split between both channels while DRW’s is almost entirely transported. Error bars are 95% Hájek intervals on $\widehat{\Delta R}$. Right: covariance gain by training-frequency tier for both schedules, showing the same head-loses/tail-gains signature as Figure 6 despite the change of domain and grouping variable.

Table 14: Covariance gain by training-frequency tier at the superclass audit, 20-Newsgroups-LT. Both schedules gain within-group covariance on rare and medium categories and lose it on frequent ones.

Tier	n	$\widehat{\Delta R}$, CB vs CE	$\widehat{\Delta R}$, DRW vs CE
Few-shot (< 20)	1954	0.04991	0.07352
Medium-shot (20–100)	2610	0.19213	0.07171
Many-shot (> 100)	2968	−0.09227	−0.10753